

The Abductive Locking Method:
A Systematic Procedure for Discovering Ontological Axioms from
Boundary Crises

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Abstract

How are first principles—the foundational axioms of metaphysics and theoretical science—discovered? Historical answers to this question have either mystified the discovery process as the private insight of genius (Descartes's intuition, Popper's conjecture) or dissolved it into arbitrary symbolic choice (Carnap's convention). This paper proposes and systematically articulates a method for first-principle discovery that is publicly testable, replicable, and revisable: the **Abductive Locking Method** (*suoyin fengsuo fa*). The Abductive Locking Method consists of five sequential operations. **(1) Pattern recognition of boundary crises.** The researcher identifies a shared deep structure across boundary predicaments in mathematics, physics, biology, and social systems—phenomena in which systems, when pushed toward operational limits such as unlimited self-reference, infinite subdivision, or total control, trigger collapse or undergo fundamental restructuring. **(2) Logical inversion.** The guiding question is inverted from "Why do systems fail at their boundaries?" to "How is it that systems manage to persist stably?" This inversion transforms stability from an assumed background into an achievement requiring explanation. **(3) Delimitation and locking of possibility space.** The researcher exhausts the logically possible positions on the target question, establishes explicit criteria, and eliminates positions that prove incoherent or explanatorily redundant, until only one self-consistent position remains. **(4) Stress testing.** The remaining position is subjected to the harshest boundary cases available—phenomena that could potentially falsify it. If it passes, it is confirmed; if it fails, the locking procedure is reopened. **(5) Formalized statement.** The surviving position is formulated as an axiom or theorem, with explicit specification of its dependencies, core proposition, elaboration, interdisciplinary anchors, and philosophical dialogue.

This paper provides an epistemological defense of the Abductive Locking Method, clarifying why it must proceed from boundary crises, why it employs abductive inference, and what legitimates the locking operation on possibility space. Through systematic dialogue with traditional methodological approaches—Descartes, Bacon, Kant, and Popper—the paper positions the Abductive Locking Method not as a negation of prior methods, but as a rational reconstruction and synthesis of what each captured only partially. The discovery of "Meta-Theorem 2: The Interactional Co-Constitution of Cognition and Being" is presented as a full-scale demonstration of the method's operation from pattern recognition to formalized statement.

The Abductive Locking Method marks a transition in first-principle discovery from "the private affair of genius" to "a publicly operable method." It does not promise that the propositions discovered are absolute truths; it promises only that, given a delimited possibility space and explicit criteria, the remaining proposition is the most coherent, parsimonious, and explanatorily powerful candidate within the current cognitive framework. If someone points out an omission in the possibility space, an unfairness in the criteria, or an oversight in stress testing, the lock is reopened. This openness is precisely the method's strength.

Keywords: Abductive Locking Method; first-principle discovery; ontological axioms; boundary crises; abductive reasoning; methodology

A Note on the Style of This Work

This work adopts a "problem-driven" genre of writing that has become uncommon in contemporary academic production. The argument does not begin with a review of the scholarly literature; it begins with a question that anyone might encounter in the course of serious reflection: How are the first principles of metaphysics and theoretical science actually discovered?

This choice is not a gesture of nostalgia for classical forms. It follows from the very theory that this work practices. According to the diagnosis of "conceptual suspension"—a term explained in the footnote below^[^1]—the "literature-driven" paradigm of contemporary academic writing tends, over generations of transmission, to let the signifieds of technical terms drift while their signifiers remain stable, thereby severing the connection between knowledge production and the cognitive needs of the public. The present work attempts a counter-demonstration: every core term, upon its first occurrence, will be anchored either to a publicly accessible experience or to an operationally defined procedure. Every step of the argument invites the reader to test it with their own reason.

In this sense, this work is not merely an exposition *about* the Abductive Locking Method; it is an enactment of the cognitive ethic that the method itself demands.

[^1]: **Conceptual suspension** (*gainian xuanfu*). A phenomenon in which the signifier (the symbolic form) of an academic concept remains stable across generations of transmission, while its signified (the empirical phenomenon or problem it originally pointed to) drifts due to the loss of the concept's unfolding path back to its experiential anchor. It is the cognitive manifestation of a drifting right-anchor within the academic community. For a full exposition of this concept, see the author's companion paper, *Academic Jargon as Critique: Alienation and Antagonism in Modern Knowledge Production* (DOI: 10.5281/zenodo.19642458).

Key Terms

This section provides concise definitions for the core terms employed in the argument that follows. Each entry includes the term, a brief definition, and an anchor—an indication of the publicly accessible experience or operation to which the term refers. These terms will also be explicated in context as they appear in the body of the text.

Abduction / Abductive Reasoning

: **Definition.** A mode of inference that begins from an observed phenomenon and works backward to the most plausible conditions that would make that phenomenon expectable. **Anchor.** A physician inferring the disease from the symptoms; a detective inferring the sequence of events from the clues. The present work systematizes abduction into a method for first-principle discovery.

Locking

: **Definition.** The operation of exhausting the logically possible positions on a given question and then eliminating those that prove incoherent or explanatorily redundant, until only one self-consistent position remains. **Anchor.** As in a locked room where one tests every door, eliminating those that cannot be opened, until only one possible exit remains—which, whatever lies beyond it, is the sole viable path.

Boundary Crisis

: **Definition.** The phenomenon, observable across disciplines, in which a system—mathematical,

physical, biological, or social—when pushed toward its operational limits (unlimited self-reference, infinite subdivision, total control), triggers collapse or undergoes fundamental restructuring. **Anchor.** Gödel's incompleteness theorems (the boundary of mathematical formalism); the ultraviolet catastrophe (the boundary of classical physics); the self-immune attack (the boundary of biological self-recognition); the paradox of totalitarian control (the boundary of social organization).

Possibility Space

: **Definition.** The set of all logically tenable positions that can be taken with respect to a precisely formulated question. **Anchor.** Before a multiple-choice question is answered, the set of all given options constitutes a miniature possibility space. The present method expands this operation to the space of fundamental philosophical alternatives.

Logical Inversion

: **Definition.** The operation of turning a guiding question from "Why does the system fail at its boundary?" to "How is it that the system manages to persist stably?"—thereby transforming stability from an assumed background into an achievement requiring explanation. **Anchor.** The Copernican inversion: instead of asking how the heavens move around the Earth, asking how the Earth moves among the heavens. The inversion does not answer the question, but opens a new space in which an answer can be sought.

Stress Test

: **Definition.** The operation of subjecting a candidate position to the harshest boundary cases available—phenomena that could potentially falsify it—in order to determine whether it remains coherent under extreme conditions. **Anchor.** An engineer testing a bridge design not under normal traffic but under the maximum conceivable load and wind conditions. If the design holds, confidence in it increases; if it fails, the design must be revised.

Formalized Statement

: **Definition.** The final formulation of a surviving position as an axiom or theorem, with explicit specification of its logical dependencies, core proposition, elaboration, interdisciplinary anchors, and philosophical dialogue. **Anchor.** Euclid's *Elements*: each proposition is stated in a fixed format (enunciation, construction, proof, conclusion) so that any trained reader can verify the reasoning without re-tracing the discoverer's path.

Pattern Recognition

: **Definition.** The cognitive operation of discerning a shared deep structure across superficially disparate phenomena. **Anchor.** A chess master glancing at a board and "seeing" the positional pattern without analyzing each piece individually; a radiologist scanning an image and "feeling" the anomaly before consciously identifying it. In the present method, pattern recognition operates across boundary crises to extract their formal homology.

Dichotomous Approximation

: **Definition.** The actual cognitive procedure underlying the ideal of "locking": the possibility space is repeatedly divided into two halves; the researcher judges, on the basis of tacit pattern-matching, which

half lies closer to the truth; the chosen half is retained and the process iterates until no further division is possible. **Anchor.** The children's game of "twenty questions," in which the answer is approached not by guessing it directly but by successively narrowing the space of possibilities. The method described here is, at its cognitive core, a disciplined version of this universal human strategy.

Tacit Knowledge

: **Definition.** Knowledge that is held in a form not readily expressible in propositional language—skills, sensitivities, and pattern-recognition abilities acquired through prolonged practice and immersion in a domain. **Anchor.** The ability to recognize a face without being able to describe the features that enable the recognition; the ability to ride a bicycle without being able to state the equations of balance. The present work explicitly acknowledges that the Abductive Locking Method, in its actual discovery-phase operation, depends irreducibly on such tacit capacities, even as it provides a formalized scaffold for public testing and teaching.

First Principles

: **Definition.** Foundational propositions that cannot be deduced from any deeper premises yet possess universal binding force within their domain—metaphysical meta-axioms, basic laws of logic, mathematical axioms, cornerstones of theoretical science. **Anchor.** Euclid's parallel postulate: it is not derived from other propositions, yet the entire edifice of Euclidean geometry rests upon it. Newton's three laws of motion: they are not proven, but all of classical mechanics follows from them. The Abductive Locking Method is designed for the discovery of precisely this kind of proposition.

Meta-Theoretical Discovery

: **Definition.** The process of discovering first principles itself. Distinguished from deduction within a theory (which moves from axioms to theorems), meta-theoretical discovery asks: "Where do the axioms themselves come from?" **Anchor.** In learning geometry, you are asked to prove that the sum of a triangle's angles is 180 degrees, starting from the parallel postulate. But have you ever asked: Where did the parallel postulate itself come from? Why did Euclid choose it? That is the question of meta-theoretical discovery.

Cognitive Framework

: **Definition.** The basic presuppositions, value priorities, and patterns of thought through which an individual or community understands the world. It determines what is noticed, what is ignored, and how phenomena are interpreted. **Anchor.** Two people read the same economic news; one sees proof of the free market's efficiency, the other sees evidence of government failure. The "facts" they perceive differ not because the facts themselves differ, but because they wear different cognitive-framework lenses. The present method demands that the researcher maintain metacognitive vigilance toward their own framework.

Self-Organization Tendency

: **Definition.** An intrinsic dynamical tendency within existence itself toward order and structure. It is the fundamental dynamic underlying the emergence of systems from chaos, life from matter, and civilization from individual interaction. **Anchor.** A vortex spontaneously forming in flowing water; nebulae condensing into stars under gravity; single-celled organisms emerging from the primordial soup. None of these are the result of external design; they are manifestations of a tendency inherent in the systems

themselves. The Abductive Locking Method is itself an emergence of this self-organization tendency within the cognitive domain.

Systemic Constraint

: **Definition.** A set of self-limiting rules necessarily embedded within any system that stably persists. It ensures the system's stable persistence by dissolving the legitimacy of operations that would otherwise lead to logical or structural collapse. **Anchor.** The law of non-contradiction—a system that permits "A and not-A simultaneously" cannot sustain any coherent discourse and annihilates itself logically. Negative selection in the immune system—T-cells that attack the body's own tissues are eliminated, preventing immunological self-destruction. These are not externally imposed regulations; they are internal conditions a system must satisfy to stably exist.

Tacit Criteria

: **Definition.** The internalized, not-fully-propositionalizable standards that a researcher relies upon, in actual practice, to judge which position "lies closer to the truth." They are the product of pattern sensitivity developed through prolonged theoretical practice. **Anchor.** An experienced art authenticator glances at a painting and immediately "senses something wrong," yet cannot immediately articulate what. That sense of wrongness is their internalized tacit criteria signaling a mismatch. The "sense of unease" that the present author experiences during theoretical discovery is precisely the operation of such tacit criteria.

Introduction

How are the first principles of metaphysics and theoretical science actually discovered?

This is a question that anyone who has received an education has encountered at some point. We learn Newtonian mechanics and are told the three laws of motion and the law of universal gravitation. We learn geometry and are told the five postulates and the entire deductive edifice of Euclidean geometry that follows from them. We learn evolutionary theory and are told that natural selection is the core mechanism of speciation. But when we press further—"How did Newton actually arrive at universal gravitation?"—the textbook answer is often the anecdote of the falling apple: an oversimplified tale of sudden genius. When we ask, "Why did Euclid choose these five postulates rather than others as his axioms?" the textbook typically falls silent.

This silence is not accidental. It reflects a disquieting fact: **we do not know how first principles are discovered.** Or more precisely, we possess no publicly testable, replicable, and teachable method for discovering them.

This is not to say that no one has attempted to answer the question. Philosophers and scientists have offered various responses, but these responses have either mystified the discovery process or dissolved the question altogether.

Descartes's answer was intuition. In the *Meditations on First Philosophy*, Descartes arrived at the *cogito ergo sum*—"I think, therefore I am"—through a process of radical doubt, treating this as an unshakeable foundation from which the entire edifice of knowledge could be deduced. But when pressed on how the *cogito* itself was discovered, Descartes appealed to "clear and distinct intuition." What is intuition? How does one judge whether an intuition is clear and distinct? When and where and by whom can such intuition be reliably obtained? Descartes offered no operational answers. His method is, at bottom, a post hoc rationalization of the private insights of genius.

Bacon's answer was induction. In the *Novum Organum*, Bacon criticized the deductive and speculative excesses of traditional philosophy and argued that knowledge should ascend gradually from systematic empirical observation to universal principles through induction. This sounds reasonable, but it faces a fatal difficulty: how can universal first principles be logically derived with necessity from finite empirical observations? Hume's problem of induction—why should we believe the sun will rise tomorrow—has never been resolved within a purely inductivist framework. Bacon's method captures one aspect of scientific discovery (empirical observation), but it fails to explain how the logical gap between observation and first principle is actually bridged.

Kant's answer was transcendental deduction. Kant inverted the question. He did not ask "How do we discover first principles from experience?" but rather "How do first principles (which he called transcendental categories) make experience possible in the first place?" In his framework, basic categories such as causality, space, and time are not derived from experience; they are inherent structures of human cognition itself, prior to and constitutive of all possible experience. This answer is profound, but it anchors the source of first principles in the "universal necessity of human cognitive structure"—a presupposition that was untestable in Kant's time and remains deeply contested today. Moreover, it cannot explain why different theoretical systems choose different axioms (Euclidean versus non-Euclidean geometry), nor can it guide us in discovering new first principles.

The Logical Positivist answer was convention. Carnap and others simply dissolved the question. First principles—logical axioms, mathematical postulates—are not "discovered," they are "stipulated." We choose a set of axioms not because they are "true" but because they are "convenient." The virtue of this answer is its honesty: it admits that we have no objective method for discovering first principles. But the cost is heavy: if first principles are merely conventions, then the foundations of science are arbitrary. Why do we stipulate the law of non-contradiction rather than its opposite? Why have different civilizations independently "stipulated" the same basic logical laws? Conventionalism cannot account for the cross-cultural binding force and universal normativity of first principles.

Popper's answer was conjecture. Popper divided the logic of scientific discovery into two sharply distinct phases: the context of discovery and the context of justification. The context of discovery—how first principles are initially proposed—Popper candidly admitted possesses no logical method; it belongs to "the private affairs of genius," a matter for psychology rather than logic. The context of justification—how first principles are empirically tested—possesses rigorous method: theories must make falsifiable predictions, and those that survive falsification attempts are tentatively retained. Popper's honesty is admirable, but his conclusion had a profound effect: twentieth-century philosophy of science largely abandoned the study of discovery methods, devoting nearly all its energy to methods of testing (falsification, confirmation, paradigm shift).

The common limitation: the mystification or dissolution of discovery. Reviewing these answers, a common pattern emerges: they either mystify the discovery of first principles as the intuition of genius (Descartes, Popper) or dissolve it into empirical induction (Bacon), transcendental structure (Kant), or arbitrary convention (Carnap). Not one provides **a set of operations that is publicly testable, replicable, teachable, and revisable** for systematically discovering first principles.

This is the gap that the present work aims to fill.

The answer proposed here is the Abductive Locking Method.

The Abductive Locking Method is not a negation or replacement of prior approaches; it is a rational reconstruction of the process of first-principle discovery that has long been mystified. It does not appeal to genius intuition, does not rely on infinite induction, does not presuppose transcendental structures, and does not dissolve the question itself. It is **a systematic operational procedure that begins from boundary crises, proceeds through abductive inference and the locking of possibility space, and culminates in the formulation of ontological axioms.**

The argument of this paper will unfold in five stages. **Chapter One** sets forth the complete five-step operational sequence of the Abductive Locking Method: pattern recognition of boundary crises, logical inversion, the delimitation and locking of possibility space, stress testing, and formalized statement. **Chapter Two** provides an epistemological defense of the method—why must it be abductive? Why must it proceed through locking? Why must it begin from boundary crises? **Chapter Three** situates the method through systematic dialogue with traditional approaches—Descartes, Bacon, Kant, and Popper—locating its distinctive theoretical contribution. **Chapter Four** presents a full-scale demonstration of the method's operation, using the discovery of "Meta-Theorem 2: The Interactional Co-Constitution of Cognition and Being" as a concrete instance from pattern recognition to formalized statement. **Chapter Five** clarifies the limits and openness of the method—it does not promise that the

propositions discovered are absolute truths, but it does promise that, given a delimited possibility space and explicit criteria, the remaining proposition is the most coherent, parsimonious, and explanatorily powerful candidate within the current cognitive framework.

Before proceeding to the formal argument, it is necessary to provide a preliminary orientation to the core terms employed in this work. These terms will be explicated as they appear in the body of the text; the reader may also consult the "Key Terms" section preceding this Introduction for a concise anchor.

Chapter One: The Abductive Locking Method—A Complete Statement

This chapter sets forth the complete five-step operational sequence of the Abductive Locking Method. These five steps are not a psychological description of how a particular genius happened to think; they constitute a **set of rational operations that is publicly testable, replicable, and revisable**. Any researcher possessing basic logical competence can, by following these five steps, attempt to discover the first principles of their own domain, using the boundary predicaments of that domain as the point of departure.

Before entering the detailed exposition of each step, it is useful to provide an overview of the method's logical arc.

The Abductive Locking Method begins from a repeatedly observed yet not-unifiedly-explained cluster of phenomena: across the seemingly unrelated domains of mathematics, physics, biology, and social systems, when a system is pushed toward its operational limits—unlimited self-reference, infinite subdivision, total control—it triggers a similar pattern of collapse or restructuring. This first step is called **pattern recognition**. Traditional explanations attribute such collapse to "limited resources" or "cognitive limitations." The Abductive Locking Method executes a crucial inversion: instead of asking "Why do systems fail at their boundaries?" it asks "How is it that systems manage to persist stably?" This step is called **logical inversion**. After the inversion, a deeper question emerges: *What conditions must a system satisfy in order to persist stably?* This question admits only a finite number of logically possible answers. We exhaust these possibilities and eliminate them one by one using explicit criteria, until only one self-consistent position remains. This step is called **the delimitation and locking of possibility space**. The remaining position is not yet a theorem. It must be subjected to the harshest boundary cases available—limit situations that might plausibly constitute counterexamples. If it passes, it is confirmed; if it fails, we return to the previous step and reexamine whether any positions or criteria were overlooked. This step is called **stress testing**. Finally, the position that survives stress testing is formulated as an axiom or theorem, with explicit specification of its dependencies, core proposition, elaboration, and interdisciplinary anchors. This step is called **formalized statement**.

The following sections unfold each of these five steps in detail.

1.1 Step One: Pattern Recognition of Boundary Crises

The starting point of the Abductive Locking Method is not conceptual analysis, nor logical deduction, but **the observation of a specific class of empirical phenomena**.

Across multiple domains—mathematics, physics, biology, and social systems—we can repeatedly observe a shared deep structure: when the operations of a system approach their logical or conceptual limits—whether through unlimited self-reference, infinite subdivision, or total control—the system triggers collapse or undergoes fundamental restructuring. We term such phenomena **boundary crises**.

Below are four classic cases drawn from different disciplines.

The self-reference predicament in mathematics. Gödel's incompleteness theorems (1931) proved that any consistent formal system containing elementary arithmetic contains a true proposition that cannot be proved within that system; moreover, the system cannot prove its own consistency within itself. Hilbert had dreamed of establishing a complete formal system capable of proving all true propositions about itself. Gödel demonstrated the impossibility of that dream. Earlier still, Russell discovered a paradox in naïve set theory: consider "the set of all sets that do not contain themselves." Whether such a set contains itself or not leads inexorably to contradiction. These two cases share a single structure: when a mathematical system attempts **unlimited self-reference**—proving its own consistency, defining "the set of all sets"—the system triggers logical collapse.

The infinities catastrophe in physics. At the end of the nineteenth century, classical electromagnetic theory encountered the "ultraviolet catastrophe" in explaining blackbody radiation: according to the Rayleigh-Jeans formula, the energy density of radiation tended toward infinity at short wavelengths, flatly contradicting experimental observation. The resolution (Planck's quantum hypothesis) came not by refining the calculations but by **prohibiting** the infinite subdivision of energy—introducing the concept of energy quanta and thereby fundamentally restructuring the rules of energy exchange. In the twentieth century, quantum field theory again encountered infinities when calculating the self-interaction of point particles (the divergence problem). Physicists developed the technique of "renormalization": systematically redefining the core parameters of the theory (mass, charge) so as to "absorb" the infinities into these definitions, thereby dissolving their legitimacy in observational predictions. These two cases share a single structure: when a physical model **permits infinitely fine operations** (infinite subdivision of energy, zero-distance interaction), the theory fails. The essence of the solution is not to eliminate the infinities but to **reconstruct the framework of the theory** so that the operations leading to infinities lose their legitimacy.

Immunological self-reference in biology. The core function of the immune system is to distinguish "self" from "non-self" and to attack the latter. But the immune system itself is part of "self." This generates a self-reference problem: how does the immune system avoid attacking itself? During T-cell development, the thymus executes an operation called "negative selection": T-cells that strongly recognize self-antigens are not merely "forbidden" from attacking—they are **directly eliminated**, induced to undergo apoptosis before they can mature. This is not a "rule-prohibition" on autoimmunity; it is a **legitimacy-dissolution** of the components that could trigger autoimmunity—they never even get the chance to become part of the immune system.

The control paradox in social systems. Historically, social systems that have attempted to eliminate all uncertainty through totalitarian control have invariably ended in collapse due to internal rigidity and the suffocation of innovation. When a control system pursues **absolute, exceptionless order**, it paradoxically strangles the flexibility and diversity that the system requires to adapt to its environment. The more absolute the control, the more fragile the system becomes. This is not accidental managerial failure; it is a systemic inevitability.

The shared deep structure. These four cases—the self-reference paradox in mathematics, the infinities of physics, the autoimmunity of biology, and the control paradox of social systems—appear disparate on

their surface. Yet they share a single deep structure: **when the operations of a system approach some form of "limitlessness" or "absoluteness"—unlimited self-reference, infinite subdivision, absolute control—the system encounters a logical or operational boundary at which it destabilizes.** At this boundary, the system either collapses or restructures its own rules so as to dissolve the legitimacy of the operations that triggered the crisis.

The task of pattern recognition is to strip away the domain-specific content and extract this formally isomorphic dynamical skeleton. This step presupposes no theory; it demands only honest observation and comparison. Any researcher can independently examine these cases and judge whether they do indeed share this deep structure. If the judgment is negative, the Abductive Locking Method terminates here. If it is affirmative, we proceed to the second step.

1.2 Step Two: Logical Inversion

Pattern recognition tells us: systems collapse at their boundaries. Traditional explanation stops here, attributing such collapse to "the finitude of resources," "the limits of cognition," or "technical inadequacy." This style of explanation tacitly assumes that "unlimited operations are possible in theory, but unrealizable in practice." This is, at bottom, an **epistemological retreat**—it fails to ask a more fundamental question.

The Abductive Locking Method executes a crucial **logical inversion**. We do not ask "Why do systems collapse at their boundaries?" We ask instead:

"If the universe did not possess an intrinsic tendency to universally suppress such logically self-destructive impulses, then why is it that the vast majority of systems we observe—from stable atoms to sustainable ecosystems—persist stably most of the time, rather than instantaneously annihilating in limitless paradox or chaos?"

This inversion shifts the center of gravity of the question. Stability is no longer a default background requiring no explanation; it becomes an **achievement** that demands explanation. The reality we observe is not a "flawed version that failed to achieve limitlessness," but rather a **specifically structured version that has already successfully filtered out logically self-destructive possibilities.**

The collapses that occur at boundaries are precisely the proof of the importance of this filtering mechanism: **it is only because, under normal circumstances, some deeper law prevents systems from sliding into these "illegitimate" states that we can observe a stable world. When collapse does occur, what we witness is precisely the failure of this deep constraining mechanism.**

The function of logical inversion is not to provide the answer, but to **open a new space of questions.** Before the inversion, the question is "Why do systems fail?"—and the answer points toward external resource constraints or cognitive limits. After the inversion, the question is "How do systems succeed?"—and the answer must point toward internal, constitutive conditions of stability.

This step can be performed by any researcher independently: select a boundary crisis case from a domain you know well. First, ask in the traditional manner: "Why did it fail?" Then, ask in the Abductive Locking manner: "Why was it able to succeed most of the time?" If the inverted question reveals something that the traditional perspective could not see, this step is complete.

1.3 Step Three: The Delimitation and Locking of Possibility Space

Logical inversion leads us to a new question: **What conditions must a system satisfy in order to persist stably?**

This question admits only a finite number of logically possible answers. The third step of the Abductive

Locking Method is to exhaust these possibilities and then eliminate them one by one using explicit criteria, until only one self-consistent position remains. This step contains three sub-steps.

First sub-step: Delimiting the possibility space. For a given question, we first exhaust all logically possible positions. The criterion for exhaustion is this: the set of positions constitutes a complete classification, such that any possible answer necessarily falls into one and only one class. The delimitation of possibility space depends on the precise formulation of the question. The more precise the question, the more finite the set of possible positions.

Second sub-step: Establishing the locking criteria. Criteria are the standards by which incoherent positions are eliminated. Criteria must satisfy three conditions: (1) **public observability**—the phenomena to which the criteria refer can be independently observed and confirmed by different researchers; (2) **fairness to competing positions**—the criteria do not presuppose the truth of any position under examination; (3) **relevance to systemic persistence**—the criteria must bear directly on the core question of "how systems persist stably."

Third sub-step: Elimination. Each position in the possibility space is tested against the criteria. The standard for elimination is not "falsity" but "incoherence" or "explanatory redundancy." A position is locked out when its core claims are shown to depend on unexamined presuppositions, when it fails to explain key systemic behaviors, or when it can be reduced to other, more parsimonious positions.

To demonstrate the concrete operation of this step, we take as an example the question of "the relation between cognition and being." The question can be precisely formulated as follows: What is the relation between human cognition (perception, thought, knowledge) and being itself (the world, reality, systems)?

Delimiting the possibility space. Setting aside variations in detail, all historical answers can be classified into four fundamental types:

- **Type A: Being determines cognition (Realism).** Being possesses a fixed structure; cognition is the passive reflection or discovery of this structure. Representatives: Aristotle, Locke, early Russell.
- **Type B: Cognition constructs being (Constructivism).** Being itself is an unknowable chaos; it is human cognition (language, categories, social consensus) that imposes order upon this chaos, thereby "constructing" what we call reality. Representatives: certain versions of postmodernism, strong social constructivism.
- **Type C: Cognition and being share a common transcendental root (Transcendental Philosophy).** The basic categories of cognition (time, space, causality) and the basic structures of being share a common source, because both derive from some higher principle (transcendental subject, absolute spirit). Representatives: Kant, Hegel.
- **Type D: Cognition and being are co-constituted in interaction (Process Realism).** Being and cognition are not independent entities; they are two inseparable sides of inter-systemic interaction. The "properties" of being are stable patterns that present themselves in interaction; cognition is the change of state that interaction induces within a particular system (such as the human mind). Representatives: positions partially approaching this include Piaget and Merleau-Ponty.

Establishing the criteria. We employ three criteria:

- **Criterion One: Cross-cultural consistency of cognition.** Do humans from different cultures and different eras, when interacting with the same types of systems (stone, water, animals), exhibit consensus on certain basic properties? If so, this indicates that these properties are not entirely the symbolic construction of specific cultures.
- **Criterion Two: Historical variability of cognition.** Does human cognition of the same type of system deepen and change as modes of interaction change (for instance, through the evolution of technical instruments)? If so, this indicates that cognition is not a passive reflection of a fixed structure.
- **Criterion Three: Ontological parsimony.** Without sacrificing explanatory power, does a theory presuppose metaphysical entities that are not publicly testable (such as "the thing-in-itself," "absolute spirit," "transcendental subject")?

Elimination.

Type A (Realism) can explain Criterion One (cross-cultural consistency), but it cannot explain Criterion Two (historical variability). If cognition were merely a passive reflection of a fixed structure, why has the scientific understanding of "stone" differed so dramatically across eras—from four elements to chemical compounds to atoms and quarks? Type A is locked out.

Type B (Constructivism) can explain Criterion Two (historical variability), but it cannot explain Criterion One (cross-cultural consistency). If everything were symbolic construction, why is stone hard in all cultures? Why is the law of non-contradiction valid across all civilizations? Constructivism can only appeal to "contingent consensus," but this weakens the binding force of "consistency." Type B is locked out.

Type C (Transcendental Philosophy) can explain both Criterion One and Criterion Two—consistency arises from the universality of transcendental categories, variability from the filling-in of empirical content. But it violates Criterion Three. Kant's "thing-in-itself," Hegel's "absolute spirit"—these are metaphysical presuppositions that cannot be publicly tested. Ockham's razor demands that we choose, given equal explanatory power, the more parsimonious theory. Type C is locked out.

Type D (Process Realism) passes all three criteria. It explains Criterion One: cross-cultural consistency arises from the stability of interactional patterns—any human in any culture interacting with stone will find it hard. It explains Criterion Two: historical variability arises because modes of interaction change with the evolution of technology, society, and symbolic systems—modern physics has a deeper understanding of stone because we have invented particle accelerators and quantum mechanics, new modes of interaction that bring stone into new interactional contexts. It satisfies Criterion Three: it presupposes no untestable entities; it describes only observable processes of interaction and the stable patterns that present themselves therein.

The locking of possibility space is complete. Type D is the sole self-consistent position.

1.4 Step Four: Stress Testing

The position that remains after locking is not yet a theorem. It must be subjected to the harshest boundary cases available. The aim of stress testing is not to "prove" the position correct, but to **attempt to falsify it**. If the position fails a boundary case, we return to Step Three and reexamine whether any positions or criteria were overlooked.

For Type D (cognition and being co-constituted in interaction), we select three stress-test cases.

Test One: The stone thought experiment. Before the emergence of linguistic symbols, could early

hominins pick up a stone, move it, and use it to crack open a fruit? If Type D holds, early hominins could, without any linguistic symbols, acquire cognition of the stone—its weight, hardness, throwability—through direct interaction with it. This cognition is neither a passive reflection of "the stone itself" (for the hominin knows nothing of atomic structure), nor a symbolic construction (for no symbolic system yet exists). It is precisely **a change of state induced in the hominin system by interaction**. Test result: from archaeological and anthropological evidence, early humans did indeed manufacture and use stone tools before the emergence of language. Type D passes the first test.

Test Two: The distinction between direct interaction and indirect measurement. What is the difference between a person directly picking up a stone (direct interaction) and a person measuring the stone with a ruler (indirect measurement)? If Type D holds, in direct interaction, cognition is the internal side of the interaction—the person experiences the weight of the stone directly through muscle tension and tactile feedback. In indirect measurement, cognition depends on an intermediary system (the ruler) interacting with the target system—the person infers properties of the stone indirectly by observing the interaction between ruler and stone. In both cases, cognition is the product of interaction; the difference lies only in whether the interaction is direct or mediated. Type D can account for both cases without introducing additional metaphysical presuppositions for indirect measurement. Test passed.

Test Three: Does quantum entanglement constitute a counterexample? In quantum entanglement, a measurement on particle A appears to "instantaneously" affect the state of particle B. Does this imply the existence of an "observation" that does not depend on interaction? If so, Type D would be falsified—for Type D asserts that all cognition (including observation) is the product of interaction. But physics confirms: entanglement cannot be used to transmit information. To confirm the state change of B, the observer must obtain the measurement result of A through a classical channel (interaction) and then compare it with the measurement result of B. The acquisition of information still depends on interaction. The phenomenon of entanglement does not overturn "cognition depends on interaction"; it merely reveals a nonlocal form of interaction. Type D passes stress testing.

If any of these three tests had constituted a genuine counterexample to Type D, the lock would have to be reopened. But upon examination, Type D remains self-consistent across all boundary cases.

1.5 Step Five: Formalized Statement

The position that passes stress testing is finally formulated as an axiom or theorem. Formalized statement is not a concealment of the discovery process, but a precise encapsulation of its outcome. It enables other researchers to use the proposition directly for subsequent deduction, or to test and revise it, without having to repeat the entire discovery process.

A formalized statement must contain the following elements: **dependencies** (from which existing axioms or theorems does this proposition derive?), **core statement** (a rigorous formulation of the central proposition), **elaboration** (unfolding the meaning of the proposition and dissolving possible misunderstandings), **interdisciplinary anchors** (corresponding evidence or echoes in physics, cognitive science, linguistics, and other fields), and **philosophical dialogue** (comparison with and positioning vis-à-vis existing philosophical stances).

Taking the formalization of Type D as an example (i.e., Meta-Theorem 2: The Interactional Co-Constitution of Cognition and Being):

Meta-Theorem 2: The Interactional Co-Constitution of Cognition and Being

Dependencies: Meta-Axiom Zero (The Principle of Self-Organization Tendency); Meta-Theorem One (The Eternal Dynamical Cycle of Being); Axiom Three (The Principle of Systemic Existence and Interaction); Theorem Three (The Theorem of the Observer's Framework Relativity).

Core Statement: Being and cognition are not independent entities; they are two inseparable sides of inter-systemic interaction. What we call the "properties" of being are stable patterns that present themselves in interaction; what we call "cognition" is the change of state that interaction induces within a particular system. Any dualistic description that separates being and cognition into "object" and "subject" is a cognitive illusion produced by the operation of symbolic encapsulation, not the constitutive mode of being itself.

Elaboration: (Detailed elaboration omitted here; see the *System Ontology* chapter on Meta-Theorem Two for the full treatment.)

Interdisciplinary Anchors: The measurement problem in quantum mechanics; embodied cognition theory; Piaget's genetic epistemology; Saussure's principle of arbitrariness.

Philosophical Dialogue: Systematic comparison with Realism, Constructivism, and Kantian transcendental philosophy, locating the distinctive contribution of this theorem.

With formalized statement complete, the full procedure of the Abductive Locking Method reaches its conclusion. At this point, Type D is no longer a "possible position"; it is a **theorem**—it has passed all tests from pattern recognition to stress testing, and it stands as the most coherent, parsimonious, and explanatorily powerful proposition within the current cognitive framework. It remains open to challenge and revision by new evidence or by a superior framework.

Chapter Summary

The Abductive Locking Method consists of a five-step operational sequence: pattern recognition of boundary crises, logical inversion, delimitation and locking of possibility space, stress testing, and formalized statement. Each step is publicly testable; each step's operational standards have been explicitly stated. The method depends on no genius intuition, presupposes no transcendental categories, and does not dissolve the question it seeks to answer. It is a rational reconstruction of first-principle discovery.

In the next chapter, we will provide an epistemological defense of this method: Why must it be abductive? Why must it proceed through locking? Why must it begin from boundary crises?

Chapter Two: Epistemological Defense of the Method

Chapter One set forth the complete five-step operational sequence of the Abductive Locking Method. A natural question follows: What legitimates this procedure? Why should it be effective in discovering first principles when other methods have failed? This chapter provides an epistemological defense of the method. The defense will center on three core questions: Why abduction? Why locking? Why begin from boundary crises? Finally, we will subject the method itself to a test of methodological self-reflexivity—was the discovery of the Abductive Locking Method itself conducted in accordance with the Abductive Locking Method?

2.1 Why Abduction?

The core inferential mode of the Abductive Locking Method is **abduction**. Abduction is neither

deduction nor induction. Deduction derives particular conclusions from universal premises; the certainty of the conclusion is guaranteed by the premises, but the truth of the premises themselves cannot be supplied by deduction. Induction generalizes from particular observations to universal regularities; yet Hume's problem of induction—how finite experience can yield necessary universal propositions—has never been resolved within a purely inductivist framework.

Abduction takes a third path. Its inferential form is:

An astonishing phenomenon P is observed.

If hypothesis H were true, then P would be a matter of course.

Therefore, there is reason to surmise that H is true.

A physician inferring a disease from symptoms, a detective inferring a crime's unfolding from clues, a geologist inferring the earth's history from rock strata—all are everyday instances of abductive reasoning. Abduction does not guarantee the certainty of its conclusion; it offers only the **best explanation**: among the currently available possibilities, H is the most coherent, most parsimonious, and most explanatorily powerful hypothesis.

Why must the discovery of first principles rely on abduction? **Because first principles cannot be deduced from any deeper premises—otherwise they would not be first principles; nor can they be strictly induced from finite experience—for the universality of first principles far exceeds the coverage of any finite set of observations.** Abduction is the only viable path: from the specific patterns of collapse at systemic boundaries (phenomenon P), we work backward to the conditions that a system must satisfy in order to persist stably (hypothesis H).

Traditional philosophical methodology has held an ambivalent attitude toward abduction. Peirce introduced the concept of abduction, but he treated it as a form of "insight"—a guessing faculty that could not be methodized. Popper went further, relegating the context of discovery entirely to "psychology" and abandoning any attempt at its logical reconstruction. The distinctive contribution of the Abductive Locking Method lies precisely in **upgrading abduction from a vague insight into an operationalizable sequence of steps**. Pattern recognition, logical inversion, space-locking, stress testing—these steps do not replace abduction; they provide a **rational scaffold** for it. They render abduction no longer dependent upon the private intuition of genius, but a publicly testable and replicable operation.

2.2 Why Locking?

Abduction supplies a candidate hypothesis, but abduction alone cannot adjudicate among multiple candidate hypotheses. If the possibility space contains several hypotheses all of which can explain phenomenon P, abduction reaches an impasse. **Locking** is introduced precisely to resolve this problem. The core operation of locking is: **exhaust the possibility space, establish criteria, and eliminate positions one by one**. It does not "prove" a given position to be true; it demonstrates that other positions are, under the given criteria, incoherent or explanatorily redundant. When all other possibilities have been locked out, the remaining position, however counterintuitive it may appear, is the sole self-consistent option within the current cognitive framework.

The legitimacy of locking depends on a crucial premise: **the set of logically possible positions on a precisely formulated question is finite**. If the possibility space were infinite, the locking operation could never be exhausted, and elimination would have no terminus. Why, then, is the possibility space finite?

The answer lies in **Meta-Axiom One: The Principle of Systemic Constraint**. This principle states that

any system that can be stably articulated and reasoned about must contain within itself a set of self-limiting meta-rules that dissolve the legitimacy of "limitless hybridization" or "vicious self-reference" that would otherwise lead to logical collapse. Applied to propositional systems, this constraint entails that for any precisely formulated question, the types of logically possible answers are finite. This is not a human convention; it is a condition for the question to be stably articulated and reasoned about at all. If a question admitted an infinite number of logically incommensurable types of answer, that question itself could not constitute a stable object of discourse—it would annihilate itself logically.

Thus, **the finitude of possibility space is not an empirical generalization drawn from induction, but a constitutive condition that any stably articulable question must satisfy.** The legitimacy of the locking operation is rooted in this ontological fact: the researcher is not searching blindly in an infinite space, but conducting an exhaustive inspection within a finite space already delimited by the logical structure of the question itself.

Locking bears a resemblance to Popperian falsification—both acknowledge that we cannot positively "prove" a universal proposition. But falsification is a method of **post hoc testing**: it tells us how, once a proposition has been proposed, to challenge it with empirical evidence. Locking is a method of **ante hoc discovery**: it tells us how, before a proposition has been formulated, to locate within the possibility space the candidate most worthy of testing. The two are complementary, not competitive.

2.3 Why Begin from Boundary Crises?

The starting point of the Abductive Locking Method is not conceptual analysis, not logical axioms, but **boundary crises**—the collapse or paradoxical phenomena that systems exhibit when pushed toward their operational limits. Why must the method begin here?

First, the boundary is where systemic rules become visible. In ordinary operation, the constitutive rules of a system are transparent. When we use language, we are not conscious of grammatical rules; when we reason, we are not conscious of the law of non-contradiction. Rules, when they function smoothly, remain invisible. It is only when a rule is violated or pushed to its limit that it manifests itself as "error" or "collapse." Boundary crises are precisely the moments when systemic rules **develop** out of invisibility. Gödel's incompleteness theorems are not the everyday operation of mathematics; they are the boundary effect triggered when a mathematical system pursues complete self-reference—and it is precisely this boundary effect that reveals the deep structure of the self-constraining rules within mathematical systems.

Second, the boundary is a fulcrum for inferring the necessary conditions of systemic persistence. If a system inevitably collapses when approaching a certain kind of operation, we can infer backward: for the system to persist stably, it must inherently prohibit or dissolve the legitimacy of that operation. The pattern of collapse at the boundary directly indicates the necessary conditions for systemic persistence. The ultraviolet catastrophe was not a failure of physics; what it pointed backward toward was the illegitimacy of "infinite subdivision of energy" in physical theory—and the discovery of quantization was precisely the positive response to this necessary condition.

Third, the boundary possesses cross-disciplinary universality. The self-reference paradox of mathematics, the infinities catastrophe of physics, the autoimmunity of biology, the control paradox of social systems—these phenomena span utterly disparate domains, yet they exhibit a formally isomorphic

deep structure. This cross-disciplinary universality strongly suggests that boundary crises are not peculiar to any single domain, but are the **universal cognitive manifestation of the very conditions of systemic existence**. To begin from the boundary is to begin from the most universal conditions of systems.

The starting points of traditional philosophical methodology are either self-evident concepts (Descartes), atomic experiences (Bacon), or transcendental categories (Kant). All these starting points presuppose some unexamined epistemological privilege—the clarity and distinctness of intuition, the immediate givenness of experience, the a priori necessity of categories. The Abductive Locking Method presupposes no such epistemological privilege. Its starting point—boundary crises—is a **publicly observable systemic behavior**. Anyone can independently examine the proof of Gödel's theorems, the calculations of the ultraviolet catastrophe, the mechanism of immunological self-tolerance. The starting point is public, and the entire procedure is therefore public.

2.4 Methodological Self-Reflexivity: Was the Discovery of This Method Itself Conducted According to the Abductive Locking Method?

Any meta-theoretical method must confront a self-referential test: How was this method itself discovered? And did its discovery follow the very steps it prescribes? If the answer is negative, the method falls into self-refutation—it demands of others a procedure it did not itself follow.

The Abductive Locking Method passes this test. Its own discovery is itself an instance of the Abductive Locking Method.

Step One: Pattern recognition of boundary crises. The discovery of the Abductive Locking Method began with the observation of boundary crises in the domain of meta-theoretical discovery—Descartes's intuition, Bacon's induction, Kant's deduction, Popper's conjecture. Each of these methods mystified or dissolved the discovery of first principles. This was not the accidental failure of one particular method; it was a **shared deep pattern**: traditional methodology, when it touched the boundary question of "how first principles are discovered," collectively retreated.

Step Two: Logical inversion. The traditional question was: "Why can we not logically discover first principles?" The Abductive Locking Method inverted the question: "If the discovery of first principles were impossible, why do we in fact possess Newtonian mechanics, Euclidean geometry, and natural selection—first principles whose status is widely acknowledged?" This inversion transformed "discovery" from an assumed impossibility into an achievement demanding explanation.

Step Three: Delimitation and locking of possibility space. For the question "How are first principles discovered?", the logically possible types of answer are finite: either they are innate (intuition, a priori), or empirical (induction), or conventional (arbitrary stipulation), or abductive (inference from phenomena). Using criteria—public testability, replicability, and avoidance of mysterious entities—the first three were locked out. Abduction remained the sole self-consistent option.

Step Four: Stress testing. Abduction itself is only an inferential mode, not yet a method. How could abduction be operationalized? Boundary crisis cases (Gödel, ultraviolet catastrophe, etc.) were used to test whether such phenomena could, via abductive reasoning, yield the necessary conditions of systemic persistence. The tests passed, and the five-step operational sequence gradually crystallized from this testing.

Step Five: Formalized statement. Chapter One of this work is precisely the completion of formalized statement.

The Abductive Locking Method, therefore, demands no condition it has not itself satisfied. It is **self-reflexively coherent**. This self-reflexivity is not a vicious circle; it is a necessary condition for any meta-theoretical method—just as the truth of the Gödel sentence is "seen" from outside the system, the legitimacy of the method is demonstrated in its application.

Chapter Summary

The epistemological defense of the Abductive Locking Method rests on the answers to three core questions. Why abduction? Because first principles can be neither deduced nor strictly induced; abduction, working backward from boundary phenomena to conditions of systemic persistence, is the only viable path. Why locking? Because the finitude of possibility space is a constitutive condition of any stably articulable question; locking upgrades abduction from vague insight to rational adjudication. Why begin from boundary crises? Because the boundary is where systemic rules become visible, where necessary conditions can be inferred, and where cross-disciplinary public observability is secured. Finally, the method passes the test of self-reflexivity: the discovery of the Abductive Locking Method itself followed the Abductive Locking Method.

In the next chapter, we situate the Abductive Locking Method in systematic dialogue with traditional approaches—Descartes, Bacon, Kant, and Popper—to further delineate its distinctive theoretical contribution.

Chapter Three: Dialogue with Traditional Methods

The Abductive Locking Method does not emerge from a vacuum. It inherits the rich legacy of philosophical inquiry into the problem of first-principle discovery, even as it seeks to transcend the limitations of that legacy. This chapter situates the Abductive Locking Method in systematic dialogue with four classical approaches: Descartes's intuition, Bacon's induction, Kant's transcendental deduction, and Popper's conjecture and falsification. The aim of these dialogues is not to repudiate the work of predecessors—each captured a genuine aspect of the discovery process—but to demonstrate how the Abductive Locking Method preserves what each captured while integrating those partial insights into a complete, publicly operable procedure.

3.1 Dialogue with Descartes: From Intuition to Pattern Recognition

The Cartesian insight. In the *Meditations on First Philosophy*, Descartes arrived, through the method of radical doubt, at the *cogito ergo sum*—"I think, therefore I am"—an Archimedean point he considered unshakeable. From this point, he sought to deduce the entire edifice of knowledge. When pressed on how the *cogito* itself was discovered, Descartes's answer was: **clear and distinct intuition**.

The genuine aspect captured. Descartes correctly perceived that the discovery of a first principle is often accompanied by a phenomenological experience of "clarity and distinctness." This experience is not the conclusion of a chain of reasoning; it is the premise from which reasoning proceeds. It presents itself to the mind directly, non-inferentially. Anyone who has experienced genuine theoretical breakthrough can attest to the reality of this intuitive experience. It is not an illusion; it is the phenomenological signal emitted by a cognitive system when it successfully grasps a deep pattern.

Why it failed as a whole. Descartes's error lay not in acknowledging intuition, but in **treating intuition as the entirety of the method**. He failed to answer—and could not answer—two fatal questions. First, how does one distinguish genuine intuition from arbitrary fancy? Second, if intuition is private, how can it serve as the foundation for public knowledge? Descartes ultimately appealed to God's veracity to guarantee the objective validity of clear and distinct ideas. This maneuver shifted the problem from epistemology to theology, and before the tribunal of secular reason, it fails to constitute a valid defense.

How the Abductive Locking Method inherits and upgrades. The Abductive Locking Method preserves the rational kernel of Cartesian intuition—the direct grasp of a deep pattern—but transforms it from an "ineffable private experience" into a publicly operable **pattern recognition**. The first step of the method (pattern recognition of boundary crises) is precisely the construction of a rational scaffold for intuition: it does not wait for intuition to descend from the heavens; it actively searches for formally isomorphic deep structures across boundary predicaments in multiple disciplines. Intuition is no longer the mysterious starting point; it is the **product** of pattern recognition. When a researcher repeatedly immerses herself in cases—Gödel's incompleteness, the ultraviolet catastrophe, immunological self-tolerance—the shared skeleton gradually "develops" into visibility. The feeling of "clarity and distinctness" that then arises is not a guarantee of truth, but an internal signal of successful pattern-matching. And it is always open to public testing: other researchers can independently examine the cases and judge whether they indeed share this structure.

3.2 Dialogue with Bacon: From Induction to Abduction

The Baconian insight. In the *Novum Organum*, Bacon criticized traditional philosophy's reliance on authority and deduction, arguing that knowledge should ascend gradually from systematic empirical observation to universal principles through the method of induction. He designed the "three tables"—the Table of Essence and Presence, the Table of Deviation, and the Table of Degrees—in an attempt to provide induction with a set of operable rules.

The genuine aspect captured. Bacon correctly perceived that the discovery of first principles cannot be divorced from the **systematic observation of empirical phenomena**. It is not a purely conceptual game of deduction; it must be anchored in the examination of how the world actually operates. The capacity of modern science to generate reliable knowledge continuously is, in large measure, an inheritance of this Baconian insight—the anchoring of theory in observable experience.

Why it failed as a whole. Bacon's error lay in supposing that **universal principles can be logically induced from experience**. Hume's problem of induction fatally exposed the gap: no finite number of white swans can logically entail that "all swans are white." Between finite experience and universal proposition lies a logical chasm that induction cannot bridge. Bacon's three tables can systematize observation, but they cannot explain why scientists, after observing a finite number of cases, are able to "leap" to universal first principles.

How the Abductive Locking Method inherits and upgrades. The Abductive Locking Method preserves Bacon's emphasis on empirical observation, but replaces the inferential mode of **induction** with **abduction**. Abduction does not claim to "prove" a universal proposition from finite experience; it works backward from astonishing boundary phenomena (such as the ultraviolet catastrophe) to the necessary conditions that a system must satisfy in order to persist stably (such as the quantization of energy). This is not an inductive leap from particulars to universals, but an abductive inference from effect to necessary condition. The locking operation on possibility space further ensures that, among the multiple logically possible necessary conditions, only the most coherent and parsimonious is retained.

Bacon's three tables are here reconstructed as **the establishment of criteria and the elimination of positions**—not the accumulation of positive instances to confirm, but the use of boundary cases to lock out incoherent options.

3.3 Dialogue with Kant: From Transcendental Categories to Interactional Co-Constitution

The Kantian insight. In the *Critique of Pure Reason*, Kant launched a "Copernican Revolution" of his own. He did not ask how our cognition conforms to objects; he asked instead how objects conform to our cognition. His answer: the human cognitive system possesses a set of transcendental categories (time, space, causality, etc.), and objects of experience can become cognizable "phenomena" only through the formative work of these categories. First principles (such as the law of causality) are not discovered from experience; they are the **a priori conditions that cognition imposes upon experience**.

The genuine aspect captured. Kant correctly perceived that first principles—at least some of them—possess a distinctive kind of necessity. They are not contingent empirical generalizations; they are constitutive conditions without which coherent experience would be impossible. The law of causality is not an empirical regularity that "has held so far"; it is a framework without which we cannot coherently think about the world. This insight explains the universal binding force of first principles far more profoundly than Baconian induction ever could.

Why it failed as a whole. Kant's error lay in anchoring these constitutive conditions in **an untestable "transcendental subject."** The transcendental subject is a philosophical postulate—posited as universal, necessary, and unchanging, yet Kant offered no empirical evidence for its existence or its structure. More fatally still, the Kantian framework cannot account for the **historical evolution and domain variability of first principles**. If causality is an unchanging structure of the transcendental subject, why do Aristotle's four causes differ so dramatically from the probabilistic causality of modern physics? Why can the parallel postulate of Euclidean geometry be revised in non-Euclidean geometries? Kant's only reply would be that those which were revised were not genuine transcendental categories after all. This renders his theory unfalsifiable and closed.

How the Abductive Locking Method inherits and upgrades. The Abductive Locking Method preserves Kant's profound insight into "constitutive conditions," but relocates their anchoring point from the "transcendental subject" to the **dynamics of inter-systemic interaction**. The method asks not "What are the a priori structures of human cognition?" but "What conditions must a system satisfy in order to persist stably?" The law of non-contradiction is valid not because it is inscribed in the a priori structure of the human mind, but because **any system capable of stable articulation must satisfy this condition**—a system that violates the law of non-contradiction annihilates itself logically. The law of causality is valid not because it is a transcendental category, but because **the self-organization tendency of being itself stably presents itself as causal patterns in inter-systemic interaction**. This account requires no postulate of a "transcendental subject"; it requires only the observation of actual patterns of interaction. Moreover, it can explain the historical evolution of first principles: as humanity's modes of interaction evolve with technology and symbolic systems, our cognition of the "conditions of existence" deepens and revises itself. The transition from Euclidean to non-Euclidean geometry is not a replacement of transcendental categories, but an upgrade in our symbolic encapsulation of the interactional patterns we call "space."

3.4 Dialogue with Popper: From Conjecture to Abductive Locking

The Popperian insight. In *The Logic of Scientific Discovery*, Popper divided the process of scientific

discovery into two sharply distinct phases: the **context of discovery** and the **context of justification**. The context of discovery—how first principles are initially proposed—Popper candidly admitted possesses no logical method; it belongs to "the private affairs of genius," a matter for psychology rather than logic. The context of justification—how first principles are empirically tested—does possess rigorous method: theories must make falsifiable predictions, and those that survive attempts at falsification are tentatively retained.

The genuine aspect captured. Popper correctly perceived that the proposal of a first principle **cannot be exhausted by a set of formalizable proof rules**. If such rules existed, the principle would not be a first principle—it could be deduced from deeper premises. Popper's honesty lay in refusing to pretend that we already possess a formalized method of discovery. His falsificationism, meanwhile, provided the clearest operational criteria yet for the testing of theories.

Why it failed as a whole. Popper's error lay not in his honesty, but in **abandoning the effort at methodization too early**. By relegating the context of discovery entirely to psychology, he effectively relinquished any rational reconstruction of the discovery process. The consequence was profound: twentieth-century philosophy of science devoted nearly all its energy to methods of testing, while maintaining a collective silence on methods of discovery. This silence left the question of "how first principles are discovered" shrouded in the mystique of genius—we could only wait for the next Newton, the next Einstein, without any systematic capacity to cultivate and optimize the power of discovery itself.

How the Abductive Locking Method inherits and upgrades. The Abductive Locking Method fully accepts Popper's premise: the discovery of first principles cannot be exhausted by a set of formalizable proof rules. But it rejects Popper's conclusion: that therefore no rational method for discovery is possible. What the Abductive Locking Method provides is not a set of **proof** rules, but a set of **search and lock** rules. It does not claim to deduce first principles from axioms; it begins from the public phenomenon of boundary crises and, through abductive inference and the locking of possibility space, systematically narrows the search until the most coherent candidate is locked in. This is like detective work: no set of rules can "prove" who the murderer is, but mature investigative methods—examining motives, checking alibis, analyzing physical evidence—can systematically exclude the innocent until only the guilty remain. The Abductive Locking Method is the "investigative method" of meta-theoretical discovery. It preserves the Popperian spirit of falsification (stress testing is precisely the attempt to falsify a candidate position), but extends it from **post hoc testing** to **ante hoc locking**. Popper left discovery to genius and testing to method; the Abductive Locking Method brings discovery itself within the purview of method, while acknowledging that method's terminus is not "proof" but "best candidate."

3.5 The Theoretical Positioning of the Abductive Locking Method

Through these four dialogues, the distinctive contour of the Abductive Locking Method comes into view. It does not repudiate Descartes's intuition, but operationalizes it as pattern recognition—not waiting for inspiration to strike, but actively searching for the development of patterns at the boundaries. It does not repudiate Bacon's induction, but reconstructs it as abduction—not generalizing from positive instances, but inferring necessary conditions from boundary cases. It does not repudiate Kant's constitutive conditions, but relocates their anchoring point from the transcendental subject to the dynamics of inter-systemic interaction—not presupposing the privilege of human cognition, but asking what conditions any system must satisfy to persist stably. It does not repudiate Popper's honesty or his dualism of discovery and testing, but brings discovery itself within the scope of method—not proof, but search and lock.

The Abductive Locking Method is therefore not a **replacement** for traditional methods, but their **systematic integration and upgrade**. It preserves the genuine aspect captured by each traditional approach, while dissolving the limitations that rendered each of those approaches, taken as a whole, ineffectual: the privacy of intuition, the logical leap of induction, the untestability of the transcendental subject, the mystification of discovery. In this sense, the Abductive Locking Method is not a new method competing with the old; it is the **unfinished methodological synthesis** toward which those traditional methods were groping.

Chapter Summary

This chapter has situated the Abductive Locking Method through systematic dialogue with Descartes, Bacon, Kant, and Popper. Each dialogue followed the same structure: a sincere statement of the predecessor's insight, an identification of the genuine aspect captured, an analysis of why the approach as a whole failed, and an account of how the Abductive Locking Method inherits and upgrades that aspect. The Abductive Locking Method is not a repudiation of the tradition; it is the completion of what the tradition initiated—the integration of partial insights into a complete, publicly testable, replicable, and revisable operational procedure.

In the next chapter, we will present a full-scale demonstration of the method's operation, using the discovery of "Meta-Theorem Two: The Interactional Co-Constitution of Cognition and Being" as a concrete instance from pattern recognition to formalized statement.

Chapter Four: Demonstration of the Method in Application—The Discovery of Meta-Theorem Two as an Example

Chapter One set forth the five-step operational sequence of the Abductive Locking Method. Chapter Two provided its epistemological defense. Chapter Three situated it in dialogue with traditional methods and delineated its theoretical contribution. All of this, however, remains discourse *about* the method. The value of a method must ultimately be validated by what it actually produces—can it genuinely discover first principles?

This chapter presents the complete discovery process of an important theorem—**Meta-Theorem Two: The Interactional Co-Constitution of Cognition and Being**—as a full-scale demonstration of the Abductive Locking Method's operation from the first step to the last. This theorem is chosen as the primary example for two reasons. First, its discovery process exhibits each step of the method with particular clarity, allowing the reader to compare the abstract statement of the method with its concrete application. Second, it addresses one of the oldest and most fundamental problems in the history of philosophy—the relation between cognition and being—thereby demonstrating that the Abductive Locking Method is effective even on the most intractable metaphysical questions.

The exposition in this chapter will follow the five-step sequence exactly as stated. The reader may compare each section with the corresponding section of Chapter One, testing whether the operation at each step conforms to the method's statement.

4.1 Step One: Pattern Recognition of Boundary Crises—The Cross-Disciplinary Manifestation of Cognitive Boundary Predicaments

The starting point of the Abductive Locking Method is the observation of a specific class of empirical phenomena. In the discovery of Meta-Theorem Two, this class of phenomena is: **the recurring**

predicaments that arise when human cognition approaches its own boundaries. These predicaments span physics, cognitive science, linguistics, and philosophy, and they exhibit a shared deep structure.

The measurement problem in physics. The standard interpretation of quantum mechanics (the Copenhagen interpretation) asserts that the act of measuring a quantum system irreversibly alters the state of that system. Before measurement, an electron has no determinate position; it is the measurement itself that causes the electron to "choose" a definite position. This discovery shattered the naïve presupposition of classical physics—that there exists an observer-independent, fully determinate objective world which measurement merely passively records. Niels Bohr's famous dictum—"It is wrong to think that the task of physics is to find out how nature *is*. Physics concerns what we can *say* about nature"—captures precisely this boundary predicament: when cognition (measurement) attempts to approach the limit of being (the quantum scale), the boundary between cognition and being blurs.

The representation predicament in cognitive science. Traditional cognitive science conceives of the mind as an information-processing system: sensory inputs arrive from the external world, the mind forms internal representations, and behavior is produced as output. But research in embodied cognition reveals that cognition is not confined to a "central processor" isolated from body and environment; it is the ongoing interaction of brain, body, and world. When we pick up a cup, our cognition of the cup—its shape, weight, texture—is not formed after the act of picking up; it is dynamically generated in the act itself. Representation is not a static internal copy of the world; it is the internal side of interactional patterns. Cognitive science, in attempting to explain "how the mind knows the world," touches a boundary: mind and world may not be two separable entities at all.

The arbitrariness paradox in linguistics. Saussure demonstrated that the relation between the signifier (sound-image) and the signified (concept) in linguistic signs is arbitrary. There is no intrinsic, necessary connection between the sound "cat" and that furry animal. Different languages use entirely different sounds to refer to the same entity, proving this arbitrariness. Yet Saussure also acknowledged that once a convention is established by a linguistic community, it becomes obligatory for the individual—you cannot arbitrarily change the signified of "cat." Here lies a paradox: the connection between sign and world is arbitrary (invented by humans), yet the referential function of the sign is stable (discovered by humans). Language is both our tool for constructing the world and the only channel through which we access it. Linguistics, in asking "how language refers to reality," touches a boundary: what exactly is the relation between the symbolic system and the rules of being?

The subject-object chasm in philosophy. Ever since Descartes proposed the *cogito*, modern philosophy has been trapped in a dilemma: how can mind (subject) know the external world (object)? The mind appears to be enclosed within its own ideas; it can directly grasp only its own ideas, not external things. From this arose the long debate between rationalism (knowledge originates in reason itself) and empiricism (knowledge originates in sensory experience). Kant attempted to bridge the gap with transcendental categories, but ultimately conceded that the "thing-in-itself" is unknowable. Philosophy, in asking "how the subject knows the object," touches a boundary: the very dichotomy of subject and object may itself be an unexamined presupposition.

The shared deep structure. These four boundary predicaments, drawn from disparate domains, appear diverse in content but share a formally isomorphic deep structure: **when human cognition attempts to approach its own foundation—the relation between cognition and being—the boundary between cognition and being blurs or even dissolves.** In physics, the act of measurement and the system measured prove inseparable. In cognitive science, mind and world co-present themselves in interaction. In linguistics, the arbitrariness of the sign and the stability of reference prove mutually dependent. In

philosophy, the subject-object dichotomy reaches an impasse.

The task of pattern recognition is to strip away the domain-specific content and extract this formally isomorphic skeleton. The discovery of Meta-Theorem Two took its departure precisely from this shared structure.

4.2 Step Two: Logical Inversion—From "How Does the Subject Know the Object?" to "How Do Cognition and Being Co-Constitute?"

Confronted with the above boundary predicaments, traditional philosophy asks: "How can the subject know the object?" or "How can our cognition conform to reality?" This question presupposes the dichotomy of subject and object—first there is an independent subject, then an independent object, and the problem is how to establish a connection between them.

The Abductive Locking Method executes a crucial logical inversion. We do not ask "How does the subject know the object?" We ask instead:

"If a system (the cognitive system) and another system (the system to be cognized) can, in their interaction, stably present the patterns we call 'cognition' and 'properties,' what existential conditions must these two systems satisfy?"

This inversion shifts the center of gravity from "the channel of knowing" to "the constitutive conditions of interaction." Before the inversion, the question is "How do we know?"—and the answer points toward cognitive capacities, methods, categories. After the inversion, the question is "How can cognition and being co-present themselves?"—and the answer must point toward the **ontological conditions that cognition and being, as two sides of inter-systemic interaction, must satisfy.**

This inversion is decisive. It implies that cognition and being may not be two independent entities requiring connection, but two sides of a single interactional process. Like the two faces of a coin—one cannot ask "How does the obverse contact the reverse?" because obverse and reverse are not two separable entities; they are the co-constituted form of a single coin. The relation between cognition and being may likewise be one of **co-constitution** rather than **connection**.

4.3 Step Three: Delimitation and Locking of Possibility Space—The Elimination of Four Positions

Logical inversion leads us to a new question: What are the logically possible positions on the relation between cognition and being?

Delimiting the possibility space. Setting aside variations in detail, all historical answers can be classified into four fundamental types:

- **Type A: Being determines cognition (Realism).** Being possesses a fixed structure; cognition is the passive reflection or discovery of this structure. The correctness of cognition depends on its correspondence with being. Representatives: Aristotle's theory of forms, Locke's representationalism, early Russell's logical atomism.
- **Type B: Cognition constructs being (Constructivism).** Being itself is an unknowable chaos; it is human cognition (language, categories, social consensus) that imposes order upon this chaos, thereby "constructing" what we call reality. There is no cognition-independent reality. Representatives: certain versions of postmodernism, strong social constructivism.
- **Type C: Cognition and being share a common transcendental root (Transcendental Philosophy).** The basic categories of cognition (time, space, causality) and the basic structures of being are homologous because both derive from some higher principle—either the transcendental subject (Kant) or absolute spirit (Hegel). Cognition can know being because the

two are unified at their root.

- **Type D: Cognition and being are co-constituted in interaction (Process Realism).** Being and cognition are not independent entities; they are two inseparable sides of inter-systemic interaction. The "properties" of being are stable patterns that present themselves in interaction; cognition is the change of state that interaction induces within a particular system. The two arise simultaneously in interaction and are inseparable.

Establishing the criteria. We employ three criteria to test these four positions. The selection of criteria must satisfy three conditions: public observability, fairness to competing positions, and relevance to systemic persistence.

- **Criterion One: Cross-cultural consistency of cognition.** Do humans from different cultures and different eras, when interacting with the same types of systems (stone, water, animals), exhibit consensus on certain basic properties (e.g., the hardness of stone, the fluidity of water, the autonomous motion of animals)? If so, this indicates that these properties are not entirely the symbolic construction of specific cultures.
- **Criterion Two: Historical variability of cognition.** Does human cognition of the same type of system deepen and change as modes of interaction change (for instance, through the evolution of technical instruments)? If so, this indicates that cognition is not a passive reflection of a fixed structure.
- **Criterion Three: Ontological parsimony.** Without sacrificing explanatory power, does a theory presuppose metaphysical entities that are not publicly testable (such as "the thing-in-itself," "absolute spirit," "transcendental subject")?

Elimination.

Elimination of Type A. Type A can explain Criterion One: cross-cultural consistency arises from the fixed structure of being itself—stone is hard in all cultures because it *is* hard. But Type A cannot explain Criterion Two: if cognition is merely passive reflection of a fixed structure, why has the scientific understanding of "stone" differed so dramatically across eras? Ancient Greeks believed stone to be composed of four elements; modern chemistry regarded it as a compound; contemporary physics sees it as an assemblage of atoms and quarks. If being has a fixed structure, why has human cognition of it undergone such profound historical change? Type A can only reply: earlier peoples had not yet discovered the truth. But this reply cannot explain why the process of discovery is so deeply constrained by historical conditions and technical tools—why was it Planck in 1900, not Aristotle in 350 BCE, who discovered quanta? Type A is locked out.

Elimination of Type B. Type B can explain Criterion Two: historical variability of cognition arises from changes in symbolic systems and social consensus. But Type B cannot explain Criterion One: if everything is symbolic construction, why is stone hard in all cultures? Why is the law of non-contradiction valid across all civilizations? Why did different civilizations independently "construct" the same basic logical laws? Type B can only appeal to "contingent consensus" or "cross-cultural diffusion," but this weakens the binding force of "consistency." More fatally, Type B cannot explain why stone hurts when it falls on one's foot. Pain is not a symbol; pain is the direct signal of bodily interaction with the world. Type B is locked out.

Elimination of Type C. Type C can explain both Criterion One and Criterion Two—consistency arises from the universality of transcendental categories, variability from the filling-in of empirical content and the evolution of scientific tools. But it violates Criterion Three. Kant's "thing-in-itself," Hegel's "absolute

spirit"—these are metaphysical presuppositions that cannot be publicly tested. We may ask: how did the transcendental subject itself come into being? If it is unchanging, why does human cognitive categories show signs of historical evolution (e.g., causality evolving from Aristotle's four causes to Hume's constant conjunction to modern probabilistic causality)? Type C must either admit that the transcendental subject evolves (thereby abandoning its "transcendental" status) or insist on its immutability (thereby conflicting with Criterion Two). Either way, Type C falls into incoherence. Type C is locked out.

Passage of Type D. Type D passes all three criteria. It explains Criterion One: cross-cultural consistency arises from the stability of interactional patterns—any human in any culture interacting with stone will find it hard, because the physical interactional patterns between human bodies and stone are cross-culturally stable. It explains Criterion Two: historical variability arises because modes of interaction change with the evolution of technology, society, and symbolic systems—modern physics has a deeper understanding of stone because we have invented particle accelerators and quantum mechanics, new modes of interaction that bring stone into new interactional contexts. It satisfies Criterion Three: it presupposes no untestable entities; it describes only observable processes of interaction and the stable patterns that present themselves therein.

The locking of possibility space is complete. Type D is the sole self-consistent position.

4.4 Step Four: Stress Testing—Examination in Boundary Cases

The position that remains after locking is not yet a theorem. It must be subjected to the harshest boundary cases available. For Type D, we select three stress-test cases.

Test One: The stone thought experiment. Before the emergence of linguistic symbols, could early hominins pick up a stone, move it, and use it to crack open a fruit? If Type D holds, early hominins could, without any linguistic symbols, acquire cognition of the stone—its weight, hardness, throwability—through direct interaction with it. This cognition is neither a passive reflection of "the stone itself" (for the hominin knows nothing of atomic structure), nor a symbolic construction (for no symbolic system yet exists). It is precisely **a change of state induced in the hominin system by interaction**. Test result: from archaeological and anthropological evidence, early humans did indeed manufacture and use stone tools before the emergence of language. Type D passes the first test.

Test Two: The distinction between direct interaction and indirect measurement. What is the difference between a person directly picking up a stone (direct interaction) and a person measuring the stone with a ruler (indirect measurement)? If Type D holds, in direct interaction, cognition is the internal side of the interaction—the person experiences the weight of the stone directly through muscle tension and tactile feedback. In indirect measurement, cognition depends on an intermediary system (the ruler) interacting with the target system—the person infers properties of the stone indirectly by observing the interaction between ruler and stone (alignment of markings). In both cases, cognition is the product of interaction; the difference lies only in whether the interaction is direct or mediated. Type D can account for both cases without introducing additional metaphysical presuppositions for indirect measurement. Test passed.

Test Three: Does quantum entanglement constitute a counterexample? In quantum entanglement, a measurement on particle A appears to "instantaneously" affect the state of particle B. Does this imply the existence of an "observation" that does not depend on interaction? If so, Type D would be falsified—for Type D asserts that all cognition (including observation) is the product of interaction. But physics confirms: entanglement cannot be used to transmit information. To confirm the state change of B, the observer must obtain the measurement result of A through a classical channel (interaction) and then

compare it with the measurement result of B. The acquisition of information still depends on interaction. Entanglement does not overturn "cognition depends on interaction"; it merely reveals a nonlocal form of interaction. Type D passes stress testing.

If any of these three tests had constituted a genuine counterexample to Type D, the lock would have to be reopened. But upon examination, Type D remains self-consistent across all boundary cases.

4.5 Step Five: Formalized Statement—The Final Formulation of Meta-Theorem Two

Type D, having passed stress testing, can now be formulated as a theorem. Formalized statement is not a concealment of the discovery process, but a precise encapsulation of its outcome.

Meta-Theorem Two: The Interactional Co-Constitution of Cognition and Being

Dependencies: Meta-Axiom Zero (The Principle of Self-Organization Tendency); Meta-Theorem One (The Eternal Dynamical Cycle of Being); Axiom Three (The Principle of Systemic Existence and Interaction); Theorem Three (The Theorem of the Observer's Framework Relativity).

Core Statement: Being and cognition are not independent entities; they are two inseparable sides of inter-systemic interaction. What we call the "properties" of being are stable patterns that present themselves in interaction; what we call "cognition" is the change of state that interaction induces within a particular system. Any dualistic description that separates being and cognition into "object" and "subject" is a cognitive illusion produced by the operation of symbolic encapsulation, not the constitutive mode of being itself.

Elaboration: (The full elaboration is provided in *System Ontology*; the core points are summarized here.) The elaboration includes: the dissolution of the subject-object dichotomy; the reinterpretation of the "properties" of being as stable patterns of interaction; the reinterpretation of cognition as the internal side of interaction; the repositioning of the function of symbolic systems—symbols are tools for encapsulating and transmitting interactional patterns, not cutters of the world; and an explanation of "why previous thinkers collectively erred"—the transparency of symbolic encapsulation generates the illusion.

Interdisciplinary Anchors: The measurement problem in quantum mechanics; embodied cognition theory; Piaget's genetic epistemology; Saussure's principle of arbitrariness.

Philosophical Dialogue: Systematic comparison with Realism, Constructivism, and Kantian transcendental philosophy, locating the distinctive contribution of this theorem.

4.6 Summary of the Demonstration

This chapter has presented the complete discovery process of Meta-Theorem Two as a step-by-step demonstration of the Abductive Locking Method:

1. **Pattern recognition:** Across the measurement problem in physics, the representation predicament in cognitive science, the arbitrariness paradox in linguistics, and the subject-object chasm in philosophy, a shared deep structure was identified—cognition, when approaching its own foundation, blurs the boundary with being.
2. **Logical inversion:** The question was inverted from "How does the subject know the object?" to "What existential conditions must cognition and being, as two sides of inter-systemic interaction, satisfy?"
3. **Delimitation and locking of possibility space:** Four logically possible positions (Realism, Constructivism, Transcendental Philosophy, Process Realism) were delimited and, using three criteria (cross-cultural consistency, historical variability, ontological parsimony), eliminated

one by one, leaving Type D as the sole self-consistent position.

4. **Stress testing:** Type D was tested against three boundary cases—the stone thought experiment, the distinction between direct and indirect interaction, and quantum entanglement. It passed all three.
5. **Formalized statement:** Type D was formulated as Meta-Theorem Two, with explicit specification of its dependencies, core statement, elaboration, interdisciplinary anchors, and philosophical dialogue.

This demonstration proves that the Abductive Locking Method is not merely a methodological proposal that spins in theoretical emptiness; it actually produced a significant metaphysical theorem—one that other methods failed to capture with clarity. The reader may compare each step of this chapter with the corresponding statement of the method in Chapter One, testing whether the operations are consistent and the reasoning rigorous. If inconsistencies or gaps are found, the method faces revision; if the test is passed, the method's efficacy gains empirical support.

Chapter Summary

This chapter has demonstrated the full operation of the Abductive Locking Method through the concrete discovery of Meta-Theorem Two. Each step strictly corresponds to the method's statement in Chapter One, and the operational standards for each step have been explicitly given. This demonstration also proves that one of the oldest problems in the history of philosophy—the relation between cognition and being—can be answered, within a given possibility space and explicit criteria, in a maximally coherent manner through the Abductive Locking Method. In the next chapter, we will clarify the limits and openness of the method, completing the final piece of the methodological puzzle.

Chapter Five: The Limits and Openness of the Method

A method that claims omnicompetence, that declares itself without limit, is not a method but a dogma. The final step of the Abductive Locking Method is not to proclaim that it has discovered absolute truth, but to state explicitly its own boundaries—under what conditions it is valid, under what conditions it fails, and how it can be revised and transcended. This demarcation of self-boundary is not an admission of defect; it is the mark of a method's maturity.

5.1 No Promise of Absolute Truth

The first principles discovered by the Abductive Locking Method are not "absolute truths." The method does not claim that the propositions it discovers are eternal, necessary, and unreviseable.

What it promises is this: given a delimited possibility space and explicit criteria, the remaining proposition is the most coherent, most parsimonious, and most explanatorily powerful candidate within the current cognitive framework.

This promise contains three qualifiers. First, **the qualifier of possibility space**—the conclusion of the Abductive Locking Method depends on the delimitation of the possibility space. If the delimitation contains omissions (a fifth position that was not considered), the locking is incomplete. Second, **the qualifier of criteria**—the conclusion depends on the selection and weighting of criteria. If the criteria themselves are shown to be unfair (biased toward a particular position) or invalid (irrelevant to systemic persistence), the elimination operation must be reexamined. Third, **the qualifier of cognitive framework**—the conclusion is the optimal solution "within the current cognitive framework." As

humanity's modes of interaction evolve, as technical instruments advance, as symbolic systems are upgraded, the current cognitive framework may itself be superseded by a superior framework. At that point, the Abductive Locking Method will be re-executed within a new possibility space.

This positioning distinguishes the Abductive Locking Method from traditional dogmatic metaphysics. Dogmatism claims to have discovered eternal truth and therefore, when confronted with counterevidence, can only either cling to its claims or collapse. The Abductive Locking Method acknowledges that its conclusions are "the best currently available candidates," and therefore, when confronted with new evidence or a superior framework, it can be revised without losing its core value—it was, in its time, the furthest boundary that reason could reach.

5.2 The Openness of the Lock: How a Lock Is Reopened

The term "locking" may invite misunderstanding—it may sound like a closure that is final and irrevocable. In fact, the locking of the Abductive Locking Method is **reversible**. It preserves interfaces for reopening at three points.

First, the expansion of possibility space. The Abductive Locking Method requires the exhaustion of possible positions. But "exhaustion" is a judgment based on current cognition. Historically, the possibility of non-Euclidean geometry was long overlooked during the dominance of the Euclidean paradigm; the possibility of quantum logic never entered the philosopher's imagination during the era of classical logic. If someone points out that, beyond the four positions delimited, there exists a fifth logically possible position, and that this position can pass the criteria tests, then the lock is reopened—the new possibility space will contain five positions, and the elimination operation will be re-executed within the expanded space.

Second, the reexamination of criteria. The criteria employed by the Abductive Locking Method can themselves become objects of testing. Why these three criteria and not others? How are the weights of the criteria to be determined? If someone argues that a criterion is unfair (it presupposes the truth of a particular position) or that a crucial criterion has been omitted (e.g., "the capacity to predict novel facts"), then the locking based on the original criteria must be reexamined. The selection of criteria is not a priori; it is itself abducted from cross-disciplinary boundary crises—the criteria are effective because they can distinguish historically successful candidates for first principles from unsuccessful ones. If new historical cases demonstrate that the criteria require revision, then the criteria should be revised.

Third, the escalation of stress testing. The fourth step of the Abductive Locking Method is stress testing—subjecting the candidate position to the harshest boundary cases available in an attempt to falsify it. But "harshest" is a judgment based on current cognition. Future scientific discoveries may supply boundary cases that are today unimaginable. If someone proposes a new boundary case, and the candidate position fails that case, then the lock is reopened—the candidate position must be revised, or the procedure must return to Step Three for re-locking.

These three interfaces together constitute the **openness** of the Abductive Locking Method. It is not a closed, one-time operation, but an **iterable cognitive operating system**. Every locking is provisional; every reopening is a continuation of the method's life. The Abductive Locking Method does not fear revision, for it is designed precisely for revision.

5.3 The Scope of the Method's Applicability

The Abductive Locking Method is not a universal method. It has a definite scope of applicability, beyond which it claims no efficacy.

Applicable: the discovery of first principles. The Abductive Locking Method is designed specifically for discovering first principles. By "first principles" we mean those propositions that cannot be deduced from deeper premises yet possess universal binding force—metaphysical meta-axioms, basic laws of logic, mathematical axioms, cornerstones of theoretical science. The common feature of these propositions is that their discovery cannot rely on deduction (for there are no deeper premises) and cannot rely on induction (for their universality far exceeds the coverage of any finite experience). The Abductive Locking Method, working backward from boundary crises to conditions of systemic persistence, is tailored precisely for this class of propositions.

Not applicable: the testing of everyday empirical propositions. The Abductive Locking Method is not applicable to questions like "Will it rain today?" Such propositions are more efficiently handled by other methods (consulting weather forecasts, observing cloud formations). The Abductive Locking Method does not seek to replace the methods of empirical science; it seeks to provide a rational tool for the domain that empirical scientific methods cannot reach—the discovery of first principles.

Not applicable: internal deduction within formal systems. The Abductive Locking Method is not applicable to the proof of mathematical theorems. Once axioms are locked in, the deduction from axioms to theorems follows rigorous proof rules. The task of the Abductive Locking Method is to **discover the axioms**, not to **prove the theorems**. It concludes with formalized statement, leaving the subsequent deductive work to the logical tools of specific disciplines.

Not applicable: fundamental disagreements about values. The Abductive Locking Method addresses first principles concerning the existential conditions of systems—factual propositions about what conditions a system must satisfy to persist stably. It does not directly address normative propositions—how we ought to live. That said, once a factual proposition about the persistence conditions of human systems is established (e.g., "individual niche optimality is the criterion of civilizational health"), it can supply an inescapable factual basis for normative discussion. But the leap from "is" to "ought" itself lies outside the operational scope of the Abductive Locking Method.

5.4 Openness as the Strength of the Method

A natural objection arises: if the conclusions of the Abductive Locking Method can always be reopened, of what use is it? The answer is: **openness is precisely what makes it stronger than closed methods.**

Closed methods—such as Descartes's intuition or Kant's transcendental deduction—have only two options when confronted with counterevidence: either deny the legitimacy of the counterevidence, or collapse as a whole. For they claim that their conclusions are "clear and distinct intuitions" or "transcendentally necessary categories"—claims that admit no revision. Once it is admitted that a "clear and distinct intuition" later turned out to be wrong, the reliability of the entire method collapses.

The openness of the Abductive Locking Method enables it to **absorb counterevidence and evolve.** When someone points out an omission in the possibility space, the Abductive Locking Method thanks them—for they have helped perfect the delimitation of space. When someone points out that a criterion is unfair, the Abductive Locking Method thanks them—for they have helped optimize the criteria. When someone proposes a new stress test, the Abductive Locking Method thanks them—for they have helped test the resilience of the candidate position. Each reopening is not a failure of the method, but an **upgrade** of the method.

In this sense, the Abductive Locking Method is not a static set of rules, but a **living cognitive protocol.** Its vitality lies not in the eternal correctness of its conclusions, but in its capacity, through continuous self-revision, to bring its conclusions ever closer to that "existential condition" itself—which we can

never fully reach, but to which we can asymptotically approach.

5.5 Methodological Self-Reflexivity Reaffirmed

In Chapter Two, we tested whether the discovery of the Abductive Locking Method itself followed the Abductive Locking Method. Before concluding this chapter, it is necessary to subject the statement of the method's limits to a further test of self-reflexivity: **does the statement of the method's limits itself fall under the limits it states?**

The answer is affirmative. The scope and limits stated in this chapter are delimited on the basis of the current cognitive framework. Future methodological research may reveal that the Abductive Locking Method can in fact play a role in domains currently deemed inapplicable, or that domains currently deemed applicable require stricter qualifications. The statement of this chapter is itself an application of the Abductive Locking Method—it works backward from cases of methodological failure in history (methods that collapsed due to overclaiming) to the conditions that a method must satisfy to persist stably (acknowledging boundaries, preserving interfaces for revision). Hence, the conclusions of this chapter are likewise "the most coherent candidates within the current cognitive framework," not "absolute truths."

This self-reflexivity is not a vicious infinite regress; it is the mark of a mature meta-theoretical method. Just as Gödel's incompleteness theorems revealed the boundaries of formal systems without causing mathematics to collapse—instead giving mathematics a clearer understanding of its own foundations—so the Abductive Locking Method's revelation of its own boundaries does not vitiate it, but endows it with a **sobriety of self-cognition** that other methods lack.

5.6 The Original Contour of the Method: Dichotomous Approximation and Tacit Criteria

The formalized statement of the Abductive Locking Method presented in the preceding chapters is an idealized logical reconstruction that the author has abstracted through examination of his own thought processes. This section honestly discloses the actual contour of the author's discovery practice, to serve as a calibration benchmark for the formalized procedure and as a point of departure for future revisions.

I. The Actual Mode of Thinking: Dichotomous Approximation

The formalized procedure describes the locking of possibility space as the exhaustive enumeration of all possible positions followed by their elimination one by one. The author's actual cognitive operation, however, follows a different course. After delimitating a possibility space, the author does not exhaustively list every position before eliminating them. Rather, the space is repeatedly divided into two halves; the author judges, on the basis of tacit pattern-matching, which half lies closer to the truth; that half is retained and the process iterates, until no further division is possible. The point that remains at the end of this dichotomous approximation is taken to be the optimal candidate within the current cognitive framework.

This procedure resembles the children's game of "twenty questions," in which the answer is approached not by guessing it directly but by successively narrowing the space of possibilities. The cognitive core of the method described in this work is, in its actual operation, a disciplined version of this universal human strategy.

II. The Internalization of Criteria: Tacit Knowledge of Pattern Gestalts

In making the judgments that guide dichotomous approximation, the author does not apply an explicit

set of criteria. Rather, he relies on a network of pattern gestalts internalized through prolonged theoretical practice. This network enables an intuitive perception of whether a given position is coherent, whether it shares the deep structure of cross-disciplinary boundary crises, and whether it is compatible with already established first principles. Such perception has traditionally been called "intuition." Within the framework of the present theory, however, it is the cognitive product of tacit pattern-matching, not a mystical revelation.

In the terms of *Noetic Ecology* (Theorem E25), a pattern gestalt is a cognitive protocol-packet composed of four elements: situational perception, response tendency, conceptual symbol, and value weight. Long immersion in theoretical practice has sedimented within the author's cognitive system a highly complex network of such gestalts—he "knows," in a tacit and bodily way, what kind of theoretical structure is self-consistent, what kind of boundary predicament is significant, and what kind of inferential chain is suspect. When confronted with a new theoretical statement or a new problem, this tacit network is automatically activated for pattern-matching. If the new information aligns strongly with the internalized gestalt network, a sense of fluency arises; if a mismatch exists—a structural conflict between the new information and the patterns one already holds—a "sense of unease" or "sense of wrongness" emerges. This sense of unease is not a mysterious oracle; it is the cognitive signal of information asymmetry: the tacit system has detected a structural discrepancy between the new information and the patterns one has already internalized.

Dichotomous approximation is precisely the operationalization of this pattern-matching in the search of possibility space. The author need not exhaust all possible positions, because the internalized gestalt network has already structured the space. At each bifurcation point, the author need only use the tacit system to judge "which side is more aligned," and then proceed along the direction of alignment until no further division is possible.

III. The Operation of Stress Testing: Searching for the Sense of Unease

What the formalized procedure calls "stress testing" manifests in actual practice as follows: the candidate proposition is brought against the boundary cases that the author most trusts—those already internalized as "touchstones"—to test whether a "sense of unease" is produced. This sense of unease is the signal of structural conflict between the candidate proposition and the internalized gestalt network. If no such unease arises, the author tentatively accepts the proposition; if unease does arise, the author returns to the step of dichotomous approximation and re-approaches the problem.

The "harshest" boundary cases are not guaranteed to be the harshest in any absolute sense; they are simply the most severe cases of which the author is currently aware, the logical vulnerabilities he is capable of identifying within his present cognitive framework. Stress testing is not a once-and-for-all proof, but the most rigorous examination that the current cognitive framework can supply.

IV. Why Formalize? Why Be Honest?

The value of the formalized reconstruction lies in rendering the method public. If the method remained at the tacit level, it would belong to the author alone, incapable of being taught, tested, or revised. The formalized procedure is the "grammar" of the method—it enables researchers who do not possess the same tacit gestalt network nonetheless to conduct a directed search according to explicit rules.

The value of honest disclosure lies in providing a calibration benchmark for the formalized procedure. The reader comes to understand where the "distortion" of the formalized procedure lies—it omits the central role of tacit criteria and the iterative character of dichotomous approximation. Future

methodological researchers who wish to optimize this method must return to this original contour to know where to begin. More importantly, honesty prevents the method from hardening into dogma: it openly acknowledges that it is not a set of absolute rules descended from the heavens, but a limited tool abstracted from a specific cognitive process.

Chapter Summary

The Abductive Locking Method does not promise that the propositions it discovers are absolute truths. It promises only that, given a delimited possibility space and explicit criteria, the remaining proposition is the most coherent, most parsimonious, and most explanatorily powerful candidate within the current cognitive framework. It preserves interfaces for reopening at three points: the expansion of possibility space, the reexamination of criteria, and the escalation of stress testing. Its scope is the discovery of first principles; it is not applicable to the testing of everyday empirical propositions, internal deduction within formal systems, or fundamental disagreements about values. Its openness is not a defect but a strength—it can absorb counterevidence and evolve, rather than collapsing in the face of counterevidence. Finally, the statement of the method's limits has itself been tested for self-reflexivity, demonstrating that the Abductive Locking Method is self-consistent at the meta-level.

At this point, the Abductive Locking Method has been presented in its entirety—from operational statement to epistemological defense, from dialogue with traditional methods to concrete demonstration, and finally to the demarcation of its boundaries and openness. In the Conclusion, we will summarize the core contributions of this method and indicate possible directions for its future development.

Conclusion

This work began from a question long left suspended: How are the first principles of metaphysics and theoretical science actually discovered?

Historical answers—Descartes's intuition, Bacon's induction, Kant's transcendental deduction, Popper's conjecture—have either mystified the discovery process as the private insight of genius or dissolved it into arbitrary symbolic convention. Not one provided an operational method that is publicly testable, replicable, and revisable. This methodological vacuum has left the discovery of first principles shrouded in the mystique of genius—we could only wait for the next Newton, the next Einstein, without any systematic capacity to understand, cultivate, and optimize the power of discovery itself.

The answer proposed in this work is the **Abductive Locking Method**—a five-step operational procedure that begins from boundary crises, proceeds through abductive inference and the locking of possibility space, and culminates in the formulation of ontological axioms.

Step One: Pattern recognition of boundary crises. Across mathematics, physics, biology, and social systems, identify the shared deep structure in boundary predicaments. This step anchors the method in publicly observable phenomena—it begins not from concepts but from observable patterns of systemic collapse.

Step Two: Logical inversion. Invert the guiding question from "Why do systems fail at their boundaries?" to "How is it that systems manage to persist stably?" This inversion transforms stability from an assumed background into an achievement requiring explanation, thereby opening a new space of inquiry.

Step Three: Delimitation and locking of possibility space. Exhaust the logically possible positions on

the target question, establish explicit criteria, and eliminate positions that prove incoherent or explanatorily redundant, until only one self-consistent position remains. This step upgrades abduction from vague insight to rational adjudication.

Step Four: Stress testing. Subject the remaining position to the harshest boundary cases available in an attempt to falsify it. If it passes, it is confirmed; if it fails, return to the previous step and reopen the lock.

Step Five: Formalized statement. Formulate the surviving position as an axiom or theorem, with explicit specification of its dependencies, core proposition, elaboration, interdisciplinary anchors, and philosophical dialogue.

The epistemological defense of the Abductive Locking Method rests on three core arguments. Why abduction? Because first principles can be neither deduced from deeper premises nor strictly induced from finite experience; abduction, working backward from boundary phenomena to conditions of systemic persistence, is the only viable path. Why locking? Because the finitude of possibility space is a constitutive condition of any stably articulable question; locking upgrades abduction from guesswork to rational adjudication. Why begin from boundary crises? Because the boundary is where systemic rules become visible, where necessary conditions can be inferred, and where cross-disciplinary public observability is secured.

Through systematic dialogue with traditional methods, the distinctive positioning of the Abductive Locking Method comes into view. It does not repudiate Descartes's intuition, but operationalizes it as pattern recognition—not waiting for inspiration to strike, but actively searching for the development of patterns at the boundaries. It does not repudiate Bacon's induction, but reconstructs it as abduction—not generalizing from positive instances, but inferring necessary conditions from boundary cases. It does not repudiate Kant's constitutive conditions, but relocates their anchoring point from the transcendental subject to the dynamics of inter-systemic interaction. It does not repudiate Popper's honesty, but brings discovery itself within the purview of method—not proof, but search and lock. The Abductive Locking Method is not a new method competing with the old; it is the unfinished methodological synthesis toward which those traditional methods were groping.

The discovery of Meta-Theorem Two—the Interactional Co-Constitution of Cognition and Being—was presented as a full-scale demonstration of the method's operation from pattern recognition to formalized statement. This demonstration proved that one of the oldest problems in the history of philosophy can be answered, within a given possibility space and explicit criteria, in a maximally coherent manner through the Abductive Locking Method.

The Abductive Locking Method does not promise that the propositions it discovers are absolute truths. It promises only that, given a delimited possibility space and explicit criteria, the remaining proposition is the most coherent, most parsimonious, and most explanatorily powerful candidate within the current cognitive framework. It preserves interfaces for reopening at three points: the expansion of possibility space, the reexamination of criteria, and the escalation of stress testing. Its openness is not a defect but a strength—it can absorb counterevidence and evolve, rather than collapsing in the face of counterevidence. In the broader theoretical landscape, the Abductive Locking Method and the four-principle convergence mechanism proposed in *How Does Philosophy Become Science?* stand in an upstream-downstream relation. The Abductive Locking Method is responsible for **generation**—extracting candidate first principles from boundary crises; the four-principle convergence mechanism is responsible for **filtering**—selecting, from among the candidates, those that satisfy the conditions of scientific legitimacy. Together they constitute a complete cognitive production process from the potentiality field Φ to the truth subset Σ . This division of labor provides, for the first time, a complete rational picture of "how knowledge is

possible"—from discovery to testing.

The writing of this work is itself an enactment of the cognitive ethic that the Abductive Locking Method demands. It does not begin with a review of the scholarly literature; it begins with a question that anyone might encounter. It refuses to let terms float as jargon accessible only to insiders; it anchors every core term in publicly accessible experience or operation. It invites the reader to test every step of the argument with their own reason. In this sense, this work is not merely an exposition *about* the Abductive Locking Method; it is the method's self-realization in the practice of writing.

The Abductive Locking Method inaugurates an "age of method." In this age, the discovery of first principles is no longer the privilege of a few geniuses, but a public skill that any researcher of basic rational competence can learn, use, and optimize. This does not mean that everyone will become a Newton or an Einstein—just as everyone can learn scientific method, but not everyone becomes a Nobel laureate. It does mean that those who possess the gift of discovery need no longer grope alone in the dark; they can stand on the rational scaffold provided by the Abductive Locking Method and see further than their predecessors.

Future research directions are open. Can the Abductive Locking Method be applied to the discovery of first principles in ethics? Can it be applied to the construction of foundational axioms in economics or political science? Can it be integrated with automated theorem discovery in artificial intelligence to develop computational tools that assist human researchers in abductive locking? These questions lie beyond the scope of this work, but they indicate possible trajectories for the method's future extension. The argument of this work closes here. But the conversation it opens has only just begun.

Extended Reading and Academic Background Notes

This section provides academic context, literature references, technical details, and supplementary arguments for readers seeking deeper engagement. The notes are organized thematically; readers may consult them selectively according to their interests.

Note 1: On the Full Exposition of "Conceptual Suspension"

The term "conceptual suspension" invoked in the "Note on the Style of This Work" is systematically developed in the author's companion paper, *Academic Jargon as Critique: Alienation and Antagonism in Modern Knowledge Production*. Its core thesis is this: across generations of academic transmission, the "literature-driven" paradigm of writing severs the unfolding path of technical terms. The signifier (symbolic form) remains stable, while the signified (the empirical phenomenon or problem originally pointed to) drifts, eventually severing the connection between knowledge production and the cognitive needs of the public. That paper provides a dynamical explanation of this phenomenon grounded in the "right-anchor drift" within the framework of system ontology, and it proposes institutional interfaces for reconstructing the cognitive contract between the academic community and the public. Interested readers may consult: He Guorui, *Academic Jargon as Critique: Alienation and Antagonism in Modern Knowledge Production*, DOI: 10.5281/zenodo.19642458.

Note 2: On the Intellectual History of "Abduction"

Abduction (also translated as "retroduction" or "inference to the best explanation"), as a mode of inference distinct from deduction and induction, was first proposed by the American pragmatist philosopher Charles Sanders Peirce (1839–1914) in the latter half of the nineteenth century. Peirce's classic schema for abduction is:

An astonishing fact C is observed.

If A were true, C would be a matter of course.

Hence, there is reason to suspect that A is true.

Peirce's own attitude toward abduction evolved over time. In his early work, he treated abduction as a form of "insight"—a guessing faculty not susceptible to rule-governed articulation. In his later work, he attempted to provide a logical form for abduction, but never arrived at a fully operational method. The "Abductive Locking Method" presented in this work inherits Peirce's abductive spirit—inferring backward from phenomena to the most plausible explanation—but upgrades it from a vague "insight" into a five-step operational sequence. The crucial upgrade lies here: Peirce's abduction terminates at "proposing a hypothesis," whereas the Abductive Locking Method, through the "locking of possibility space" and "stress testing," advances abduction from guesswork to rational adjudication.

Classic texts on Peirce's theory of abduction include: Peirce, C. S. (1931–1958). *Collected Papers of Charles Sanders Peirce*, Vols. 1–8. Harvard University Press. See especially the discussions of abduction in Volumes 5 and 7. For secondary literature, see: Fann, K. T. (1970). *Peirce's Theory of Abduction*. Martinus Nijhoff.

Note 3: Supplementary Cross-Disciplinary Cases of Boundary Crises

Section 1.1 of the main text employed Gödel's incompleteness, the ultraviolet catastrophe, immunological self-tolerance, and the control paradox as primary cases. Several equally significant cases of boundary crises are supplemented here for the interested reader.

Russell's Paradox (Foundations of Mathematics). In 1901, Bertrand Russell discovered a fatal paradox in Frege's *Basic Laws of Arithmetic*, then about to be published. Let R be "the set of all sets that do not contain themselves." If R contains itself, then by definition it should not contain itself; if R does not contain itself, then by definition it should contain itself. This paradox directly destroyed Frege's grand program of reducing mathematics to logic and compelled the axiomatic reconstruction of set theory (the ZF axiom system). The pattern-recognition value of Russell's Paradox lies here: when a set-theoretic system attempts **unlimited self-reference** (defining "the set of all sets"), it triggers logical collapse. The solution (ZF axioms) consists precisely in dissolving the legitimacy of this self-referential construction through **rule-prohibition**.

The Halting Problem (Boundary of Computability Theory). In 1936, Alan Turing proved that there exists no universal algorithm capable of deciding, for an arbitrary program and arbitrary input, whether that program will eventually halt. This result is structurally isomorphic to Gödel's incompleteness theorems—both arise from the self-reference predicament that emerges when a system attempts a global determination of its own properties. The Halting Problem demarcates an absolute boundary for computability theory: computability itself has limits. This is not a defect of computers, but a necessary condition that any sufficiently powerful computational system must satisfy.

Bénard Convection and Dissipative Structures (Boundary of Self-Organization). Ilya Prigogine's theory of dissipative structures revealed that open systems, when driven far from thermodynamic equilibrium, can spontaneously form spatiotemporally ordered structures. Bénard convection (heating a thin liquid layer from below, which spontaneously forms hexagonal convection cells when the temperature gradient exceeds a critical threshold) is the classic instance. The boundary-crisis value of this phenomenon lies here: it challenges a simplistic reading of the second law of thermodynamics—entropy increase is not the sole tendency of existence. Under specific conditions, systems spontaneously "counter-entropically" generate order. Meta-Axiom Zero (The Principle of Self-Organization Tendency) of the present work is abduced precisely from boundary phenomena of this kind.

For detailed analyses of these cases, see: He Guorui, *System Ontology: The Ultimate Jurisprudence of Being*, Introduction and Chapter One. DOI: 10.5281/zenodo.19353224.

Note 4: On the Logical Foundations of "Possibility Space Locking"

Section 2.2 of the main text argued that the legitimacy of possibility space locking depends on the premise that "the possibility space is finite." A more detailed logical exposition is provided here.

Why is the possibility space finite? This is not an empirical generalization derived from induction, but an ontological proposition deduced from **Meta-Axiom One: The Principle of Systemic Constraint**. Meta-Axiom One states that any system that can be stably articulated and reasoned about must contain within itself a set of self-limiting meta-rules that dissolve the legitimacy of "limitless hybridization" or "vicious self-reference" that would otherwise lead to logical collapse. Apply this principle to propositional systems: if a question admitted an infinite number of logically incommensurable types of answer, then that question itself would constitute a "limitless hybridization"—any attempt to articulate and reason about it could yield no determinate conclusion, and the question could not constitute a stable object of discourse at all.

Hence, **for any question that we can stably pose and discuss, the set of logically possible types of answer is necessarily finite**. This is not a matter of human convention; it is a constitutive condition for the question to be stably articulable in the first place. The operation of "exhausting possible positions" in the Abductive Locking Method rests on this ontological fact—the researcher is not searching blindly in

an infinite space, but conducting an exhaustive inspection within a finite space already delimited by the logical structure of the question itself.

For the complete formalized argument grounding this claim, see the chapter on Meta-Axioms and Theorem Two (The Theorem of Finite Hierarchical Nesting) in *System Ontology*.

Note 5: Cognitive Science Support for the "Stone Thought Experiment"

The "stone thought experiment" employed in Section 4.4—could early hominins pick up and use a stone before the emergence of linguistic symbols?—has robust empirical support in contemporary cognitive science.

Embodied Cognition. Research in embodied cognition demonstrates that cognition is not merely abstract symbol manipulation occurring within the brain, but the ongoing interaction of brain, body, and environment. When we pick up an object, our cognition of its "weight" is not a representation formed in the brain after the fact; it is the integration, during the act itself, of muscle tension, tactile feedback, and proprioception. Cognition is the internal side of interaction.

Piaget's Genetic Epistemology. Jean Piaget, through systematic observation of children's cognitive development, showed that basic cognitive categories (space, time, causality) are not a priori givens, but are progressively constructed through the child's sensorimotor interactions with the environment. Infants, through grasping, pushing, pulling, and throwing, abstract from these interactions the "permanence" of objects, the "continuity" of space, and the "transitivity" of causality. Piaget's work provides ontogenetic evidence that cognition is the product of interaction.

Mirror Neurons and Action Understanding. Neuroscience has discovered "mirror neurons" in macaques and humans—neurons that activate both when an individual performs an action (e.g., grasping) and when the individual observes another performing the same action. This suggests that "understanding" another's action is not a matter of symbolic inference, but an internal simulation of one's own bodily action patterns. Understanding, at the ontological level, is the internalization of interactional patterns.

Relevant classic literature: Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press. Piaget, J. (1954). *The Construction of Reality in the Child*. Basic Books. Rizzolatti, G., & Craighero, L. (2004). The mirror-neuron system. *Annual Review of Neuroscience*, 27, 169–192.

Note 6: On "Quantum Entanglement" and the Interaction Principle—A Physical Clarification

Section 4.4 employed quantum entanglement as a stress-test case for the proposition that "all cognition depends on interaction." A more precise physical clarification is provided here.

The EPR Paradox and Bell's Inequality. In 1935, Einstein, Podolsky, and Rosen (EPR) proposed a thought experiment challenging the completeness of quantum mechanics. They argued that if two particles are in an entangled state, a measurement on one seems instantaneously to determine the state of the other, violating local realism. In 1964, John Bell derived Bell's inequality, rendering the question experimentally testable. Subsequent experiments (Aspect, 1982; Zeilinger, 2015, among others) have repeatedly confirmed the predictions of quantum mechanics and ruled out local realism.

Can Entanglement Transmit Information? No. This is the crucial point. Although the correlations between entangled particles are instantaneous (or at least superluminal under certain interpretations), this correlation cannot be used to transmit any classical information. The reason: the outcome of a measurement on entangled particle A is completely random; the observer cannot control the outcome. The observer of particle B can only discover the correlation with A's outcome by receiving, through a

classical channel (subliminal or luminal), the information about A's measurement. The acquisition of information—the change in the cognitive state of the observer—still depends on interaction through the classical channel.

Significance for the Abductive Locking Method. Quantum entanglement does not furnish a counterexample of "cognition without interaction." It reveals a nonlocal form of interaction, but cognition—the change induced in an observer system by observation—still presupposes interaction (whether local or nonlocal). Type D (cognition and being co-constituted in interaction) remains self-consistent at the quantum boundary.

Classic discussions of quantum entanglement and information transfer include: Bell, J. S. (1964). On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1(3), 195–200. Aspect, A., Grangier, P., & Roger, G. (1982). Experimental realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment: A new violation of Bell's inequalities. *Physical Review Letters*, 49(2), 91–94.

Note 7: Methodological Reflection on "Pattern Recognition Sensitivity"

The discovery process of the present work itself exhibits a noteworthy cognitive feature. In dialogue, the author repeatedly displayed a heightened sensitivity to "mismatch"—when a theoretical statement failed to align with a phenomenal pattern, the author would immediately sense a "wrongness" or "unease," even if unable to articulate the precise nature of the problem. It was this sensitivity that drove the progression of the Abductive Locking Method at each step.

This sensitivity has traditionally been called "intuition" or "insight." But the practice of the Abductive Locking Method suggests that it is not a mystical gift. It is a **pattern-recognition capacity** developed through repeated immersion in cases and theoretical dialogue. The researcher extensively engages with cases of cross-disciplinary boundary crises (Gödel's incompleteness, the ultraviolet catastrophe, Russell's paradox, etc.), repeatedly discerning the shared deep structure across these cases, until this structure is internalized within the cognitive system as a "default expectation." When a new theoretical statement fails to match this internalized structure, the cognitive system emits the signal of "unease."

A question worthy of further exploration is whether this pattern-recognition sensitivity can itself be methodized. The author's own theoretical discovery process employed not only abduction but also induction and deduction. For instance, perceiving universal regularities directly from a mass of particular cases (induction), or deriving specific consequences from already established axioms (deduction). Might this imply that, just as abduction can be reconstructed as the Abductive Locking Method, induction—when placed within the context of a delimited possibility space—could also be reconstructed as a directed method of pattern verification? This question lies beyond the scope of the present work, but it indicates a possible direction for future methodological research: **the coordinated methodization of multiple inferential modes.**

On the cognitive mechanisms of expert intuition and pattern recognition, see: Klein, G. (1998). *Sources of Power: How People Make Decisions*. MIT Press. This work studies how fireground commanders, emergency room nurses, and other experts make rapid, accurate judgments under high-stakes conditions, revealing that expert intuition is essentially pattern-matching capacity developed through extensive experience. Dreyfus, H. L., & Dreyfus, S. E. (1986). *Mind over Machine: The Power of Human Intuition and Expertise in the Era of the Computer*. Free Press. This work proposes a five-stage model of skill acquisition, describing the cognitive leap from rule-following to intuitive grasp.

Note 8: On the Relation to *How Does Philosophy Become Science?*

The Conclusion of the main text noted the upstream-downstream relation between the Abductive Locking Method and the four-principle convergence mechanism. A more detailed account is provided here.

How Does Philosophy Become Science?—A Formalized Foundational Mechanism Based on System Ontology (another paper by the author) proposes a four-principle filtering mechanism through which philosophical speculation converges into scientific theory: logical legitimacy (non-contradiction), observational legitimacy (existence of a codable interaction channel with the cognitive framework), formal legitimacy (structurable, unambiguous expression), and boundary legitimacy (no illegitimate infinite nesting, with empirical upper and lower bounds). That paper defines philosophy as the potentiality field Φ (all logical possibilities), and science as the truth subset Σ (propositions passing the four-principle filter).

The relation between the Abductive Locking Method and the four-principle convergence mechanism is this:

- **The Abductive Locking Method is responsible for *generation*.** Beginning from cross-disciplinary boundary crises, through pattern recognition, logical inversion, space-locking, and stress testing, it locks in candidate first principles. It answers the question: How do we find the candidates most worthy of testing?
- **The four-principle convergence mechanism is responsible for *filtering*.** It subjects the candidates produced by the Abductive Locking Method to the four tests of logical, observational, formal, and boundary legitimacy. It answers the question: Among these candidates, which may enter the body of scientific knowledge?

Together, the two constitute a complete cognitive production process from the potentiality field Φ to the truth subset Σ . The Abductive Locking Method conducts efficient search within Φ ; the four principles perform legitimacy filtering at the boundary between Φ and Σ . This division of labor provides, for the first time, a complete rational picture of "how knowledge is possible"—from discovery to testing.

Interested readers may consult: He Guorui, *How Does Philosophy Become Science?—A Formalized Foundational Mechanism Based on System Ontology*. DOI: 10.5281/zenodo.19642345.

Note 9: Operational Checklist for the Abductive Locking Method

To facilitate practical application by researchers, the five-step operation of the Abductive Locking Method is here distilled into a checklist. Researchers attempting to discover first principles in their own domains may consult this checklist step by step.

Step One: Pattern Recognition of Boundary Crises

- Have at least three potential cases of boundary crises been identified across distinct disciplines?
- Do these cases all involve systems approaching operational limits (unlimited self-reference, infinite subdivision, total control, etc.)?
- Has the domain-specific content been stripped away to extract the formally isomorphic deep structure?
- Can this deep structure be independently identified and confirmed by other researchers?

Step Two: Logical Inversion

- What is the traditional way of posing the question? ("Why does the system fail?")
- What is the inverted way of posing the question? ("How does the system manage to persist?")
- Does the inverted question open a new space of inquiry?
- Does the new question point toward constitutive conditions of the system rather than external constraints?

Step Three: Delimitation and Locking of Possibility Space

- Have all logically possible positions on the target question been exhausted?
- Is the delimited possibility space complete (no omissions) and mutually exclusive (no overlaps)?
- Are the locking criteria explicitly stated? Do the criteria satisfy the three conditions of public observability, fairness, and relevance to systemic persistence?
- For each position eliminated, is there an explicit reason for elimination (incoherence or explanatory redundancy)?
- Is the remaining position unique?

Step Four: Stress Testing

- Have at least three of the harshest boundary cases available been selected?
- Could these cases plausibly constitute counterexamples to the remaining position?
- How does the remaining position perform under each test case? Does it remain self-consistent?
- If any test fails, has the procedure returned to Step Three to re-lock?

Step Five: Formalized Statement

- Have the dependencies of the proposition been explicitly specified (from which existing axioms or theorems does it derive)?
- Is the core statement rigorously formulated and unambiguous?
- Has the elaboration sufficiently unfolded the meaning and dissolved possible misunderstandings?
- Have interdisciplinary anchors been provided (corresponding evidence or echoes in physics, cognitive science, linguistics, etc.)?
- Has philosophical dialogue been conducted (comparison with and positioning vis-à-vis existing stances)?

Bibliographic Guidance

The core literature referenced in this work has been distributed across the notes above. A systematic guide to further reading, organized thematically, is provided here.

Abduction and Methodology of Discovery:

- Peirce, C. S. (1931–1958). *Collected Papers of Charles Sanders Peirce*, Vols. 1–8. Harvard University Press.
- Fann, K. T. (1970). *Peirce's Theory of Abduction*. Martinus Nijhoff.
- Hanson, N. R. (1958). *Patterns of Discovery*. Cambridge University Press. (Proposes the "theory-ladenness of observation" and emphasizes abductive reasoning in scientific discovery.)
- Popper, K. (1934/2002). *The Logic of Scientific Discovery*. Routledge. (The classic of falsificationism; adopts a pessimistic stance on the logic of discovery.)

System Ontology and Meta-Theory:

- He Guorui. (2026). *System Ontology: The Ultimate Jurisprudence of Being*. DOI: 10.5281/zenodo.19353224.
- He Guorui. (2026). *Noetic Ecology: A Systems Theory of Cognition, Culture, and Society*. DOI: 10.5281/zenodo.19353340.
- He Guorui. (2026). *Civilization Dynamics: Planetary Civilizology*. DOI: 10.5281/zenodo.19352998.

Cross-Disciplinary Cases of Boundary Crises:

- Gödel, K. (1931/1992). *On Formally Undecidable Propositions of Principia Mathematica and Related Systems*. Dover.

- Turing, A. M. (1936). On computable numbers, with an application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society*, 2(42), 230–265.
- Prigogine, I., & Stengers, I. (1984). *Order out of Chaos: Man's New Dialogue with Nature*. Bantam.
- Wilson, K. G., & Kogut, J. (1974). The renormalization group and the ϵ expansion. *Physics Reports*, 12(2), 75–199.

Cognitive Science and Embodied Cognition:

- Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press.
- Piaget, J. (1954). *The Construction of Reality in the Child*. Basic Books.
- Clark, A. (1997). *Being There: Putting Brain, Body, and World Together Again*. MIT Press.

Foundations of Quantum Mechanics:

- Bell, J. S. (2004). *Speakable and Unspeakable in Quantum Mechanics* (2nd ed.). Cambridge University Press.
- Zeilinger, A. (2010). *Dance of the Photons: From Einstein to Quantum Teleportation*. Farrar, Straus and Giroux.